Lecture 13: Nature Language Generation

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Overview

Today we'll be learning about some approaches to **Natural Language Generation (NLG)**

Plan for today:

- Background: LMs and decoding algorithms
- More on decoding algorithms
- NLG tasks and approaches to them
- NLG evaluation: a tricky situation
- Current trends, and the future
- ChatGPT

Background: LMs and decoding algorithms

Natural Language Generation (NLG)

- Natural Language Generation refers to any setting in which we generate (i.e. write) new text.

- NLG is a subcomponent of:
  - Machine Translation
  - (Abstractive) Summarization
  - Dialogue (chit-chat and task-based)
  - Creative writing: storytelling, poetry-generation
  - Freeform Question Answering (i.e. answer is generated, not extracted from text or knowledge base)
  - Image captioning
  - …

Language Modeling

- **Language Modeling**: the task of predicting the next word, given the words so far

$$P(y_t | y_{1,\dots,}y_{t-1})$$

- A system that produces this probability distribution is called a **Language Model**

- If that system is an RNN, it's called a RNN-LM

Conditional Language Modeling

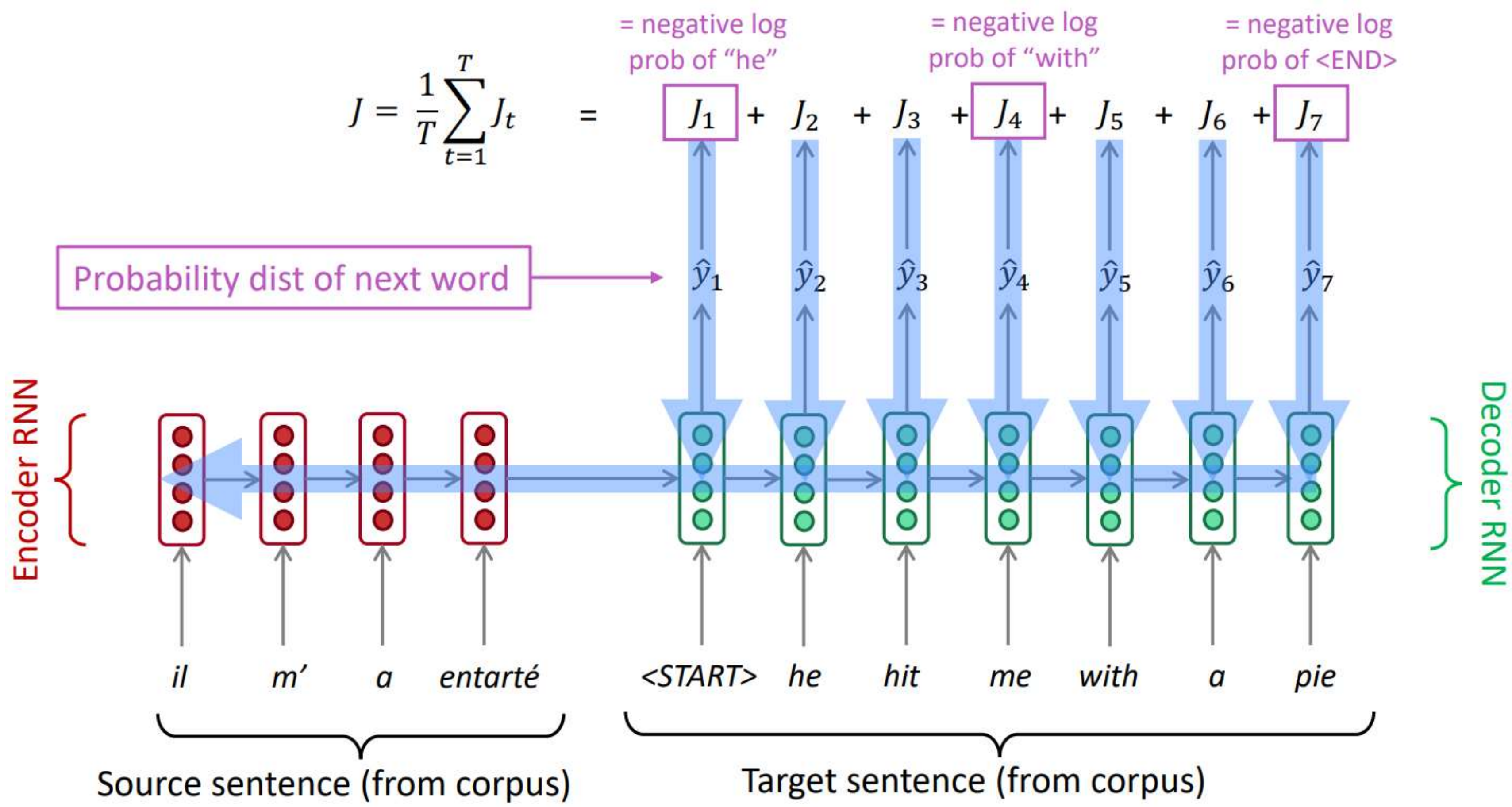
- **Conditional** Language Modeling: the task of predicting the next word, given the words so far, and also some other input x

$$P\left(y_t \middle| y_{1,\dots,}y_{t-1}, x\right)$$

- Examples of conditional language modeling tasks:
  - Machine Translation (x=source sentence, y=target sentence)
  - Summarization (x=input text, y=summarized text)
  - Dialogue (x=dialogue history, y=next utterance)
  - …

Training a (conditional) RNN-LM

- This example: Neural Machine Translation



During training, we feed the gold (aka reference) target sentence into the decoder, regardless of what the decoder predicts. This training method is called Teacher Forcing.

Decoding algorithms

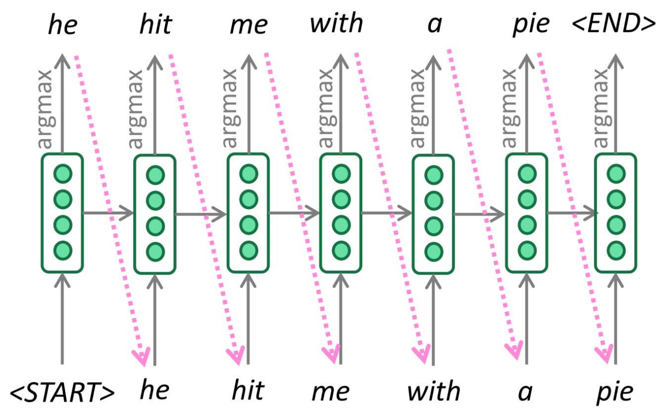
- Question: Once you've trained your (conditional) language model, how do you use it to generate text?

- Answer: A decoding algorithm is an algorithm you use to generate text from your language model

- Widely used two decoding algorithms:
  - Greedy decoding
  - Beam search

Greedy decoding

- A simple algorithm
- On each step, take the most probable word (i.e. argmax)
- Use that as the next word, and feed it as input on the next step
- Keep going until you produce <END> (or reach some max length)



Beam search decoding

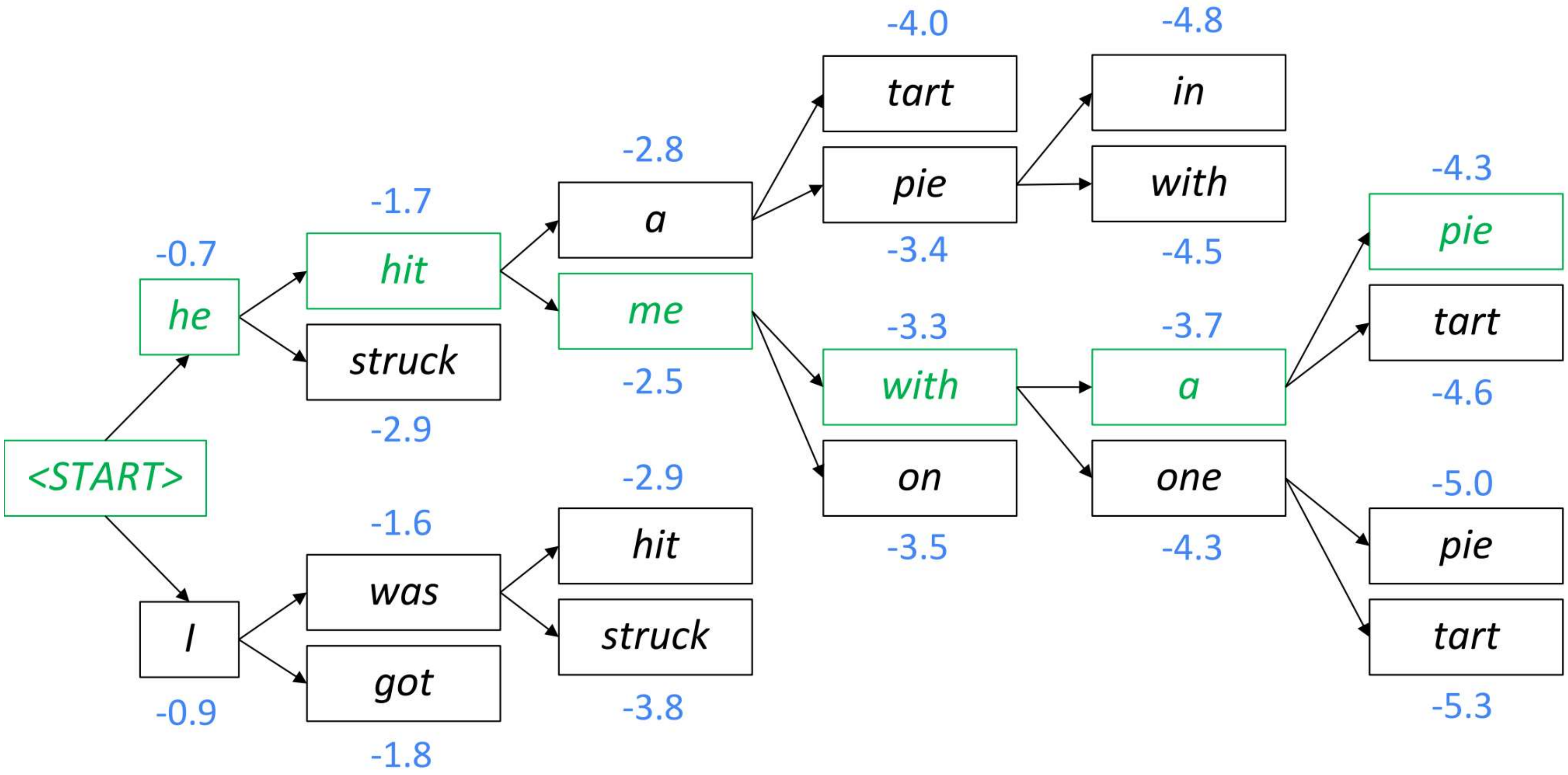
- A search algorithm which aims to find a high-probability sequence (not necessarily the optimal sequence, though) by tracking multiple possible sequences at once.

- Core idea: On each step of decoder, keep track of the k most probable partial sequences (which we call hypotheses)
  - k is the beam size

- After you reach some stopping criterion, choose the sequence with the highest probability (factoring in some adjustment for length)

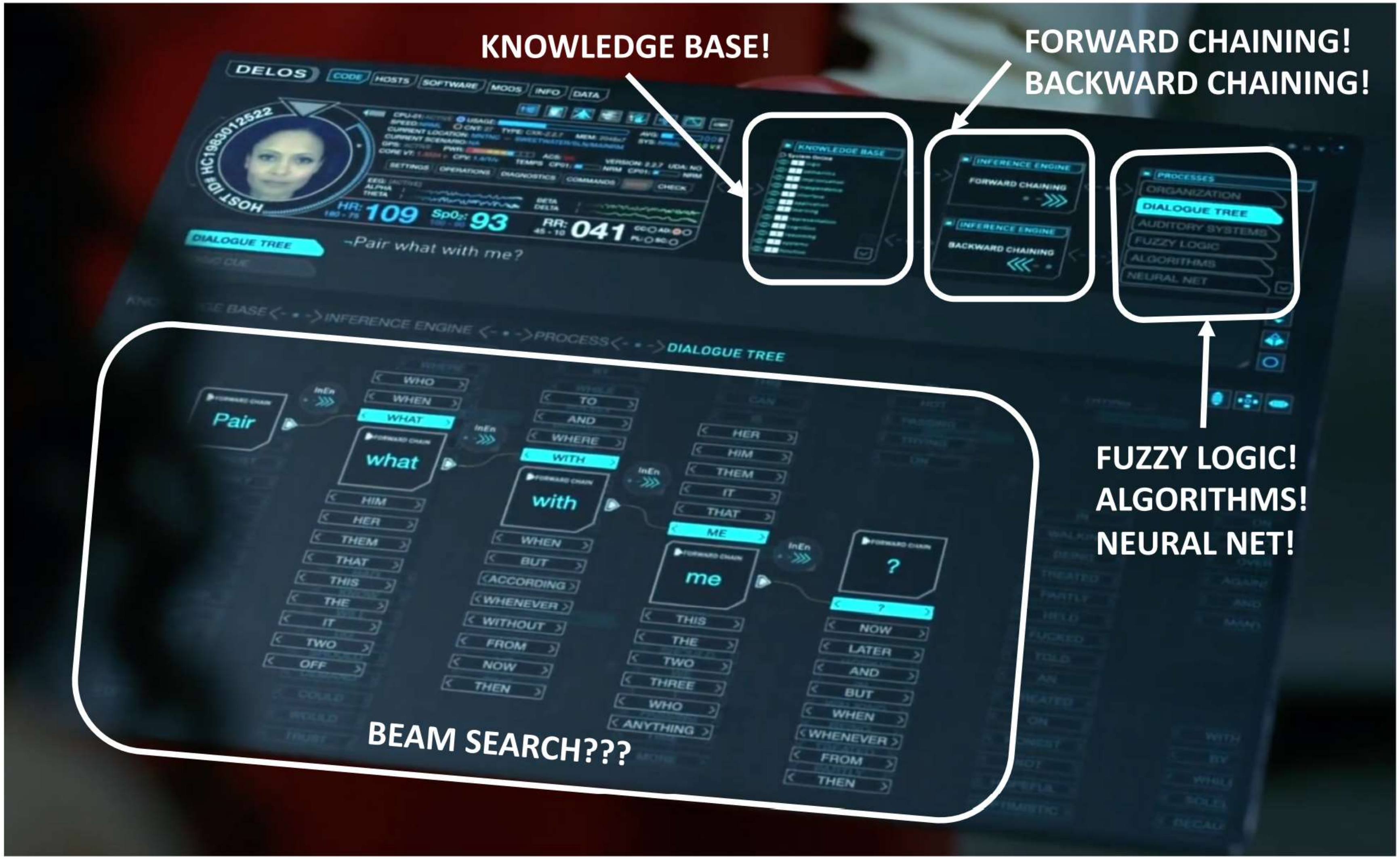
Beam search decoding

- Beam size = k = 2
- Blue numbers = $\text{score}(y_1, \ldots, y_t) = \sum_{i=1}^{t} \log P_{\text{LM}}(y_i | y_1, \ldots, y_{i-1}, x)$



Westworld（《西部世界》）

- Do the hosts in Westworld use beam search?



Beam search decoding

- What's the effect of changing beam size k?

- Small k has similar problems to greedy decoding (k=1)
  - Ungrammatical, unnatural, nonsensical, incorrect

- Larger k means you consider more hypotheses
  - Increasing k reduces some of the problems above
  - Larger k is more computationally expensive
  - But increasing k can introduce other problems:
    - For NMT, increasing k too much decreases BLEU score (Tu et al, Koehn et al). This is primarily because large-k beam search produces too short translations (even with score normalization!)
    - In open-ended tasks like chit-chat dialogue, large k can make output more generic

Effect of beam size in dialogue

I mostly eat a fresh and raw diet, so I save on groceries

Human chit-chat partner

Beam size	Model response
1	*I love to eat healthy and eat healthy*
2	*That is a good thing to have*
3	*I am a nurse so I do not eat raw food*
4	*I am a nurse so I am a nurse*
5	*Do you have any hobbies?*
6	*What do you do for a living?*
7	*What do you do for a living?*
8	*What do you do for a living?*

Low beam size:
More on-topic but nonsensical;
bad English

High beam size:
Converges to safe, "correct" response, but it's generic and less relevant

Sampling-based decoding

- Pure sampling
  - On each step t, randomly sample from the probability distribution Pt to obtain your next word.
  - Like greedy decoding, but sample instead of argmax.

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  - On each step t, randomly sample from P_t, restricted to just the top-n most probable words
  - Like pure sampling, but truncate the probability distribution
  - n=1 is greedy search, n=V is pure sampling
  - Increase n to get more diverse/risky output
  - Decrease n to get more generic/safe output

Sampling-based decoding

Both of these are **more efficient** than beam search – no multiple hypotheses

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Softmax temperature

- Recall: on timestep t, the LM computes a prob dist P_t by applying the softmax function to a vector of scores $s \in R^{|V|}$

$$P_t(w) = \frac{\exp(s_w)}{\sum_{w' \in V} \exp(s_{w'})}$$

- You can apply a temperature hyperparameter τ to the softmax:

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  - Thus more diverse output (probability is spread around vocab)
- Lower the temperature τ: P_t becomes more spiky
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Note: softmax temperature is not a decoding algorithm!
It's a technique you can apply at test time, in conjunction with a decoding algorithm (such as beam search or sampling)

Decoding algorithms: Summary

- **Greedy decoding** is a simple method; gives low quality output
- **Beam search** (especially with high beam size) searches for high probability output
  - Delivers better quality than greedy, but if beam size is too high, can return high-probability but unsuitable output (e.g. generic, short)
- **Sampling methods** are a way to get more diversity and randomness
  - Good for open-ended / creative generation (poetry, stories)
  - Top-n sampling allows you to control diversity
- **Softmax temperature** is another way to control diversity
  - It's not a decoding algorithm! It's a technique that can be applied alongside any decoding algorithm.

NLG tasks and approaches to them

Summarization: task definition

Task: given input text x, write a summary y which is shorter and contains the main information of x.

Summarization can be single-document or multi-document.

- Single-document means we write a summary y of a single document x.
- Multi-document means we write a summary y of multiple documents $x_1, \ldots, x_n$

Typically $x_1, \ldots, x_n$ have overlapping content: e.g. news articles about the same event

Summarization: task definition

Within single-document summarization, there are datasets with source documents of different lengths and styles:

- Gigaword: first one or two sentences of a news article → headline (aka sentence compression)
- LCSTS (Chinese microblogging): paragraph → sentence summary
- NYT, CNN/DailyMail: news article → (multi)sentence summary
- Wikihow: full how-to article → summary sentences

Sentence simplification is a different but related task: rewrite the source text in a simpler (sometimes shorter) way

- Simple Wikipedia: standard Wikipedia sentence → simple version
- Newsela: news article → version written for children

Summarization: task definition

Extractive summarization

Select parts (typically sentences) of the original text to form a summary.

- Easier
- Restrictive (no paraphrasing)

Abstractive summarization

Generate new text using natural language generation techniques.

- More difficult
- More flexible

Pre-neural summarization

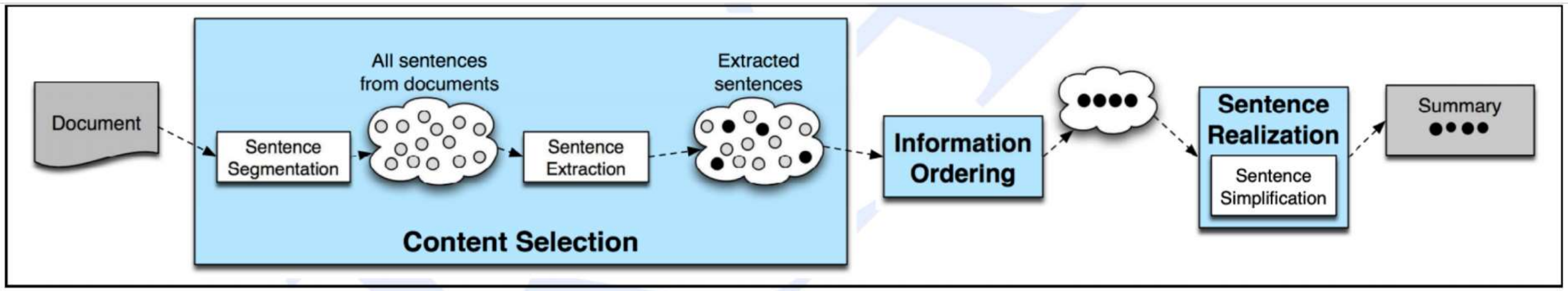


Figure 23.14 The basic architecture of a generic single document summarizer.

Pre-neural summarization systems were mostly extractive

Like pre-neural MT, they typically had a pipeline:

- **Content selection**: choose some sentences to include
- **Information ordering**: choose an ordering of those sentences
- **Sentence realization**: edit the sequence of sentences (e.g. simplify, remove parts, fix continuity issues)

Pre-neural summarization approaches

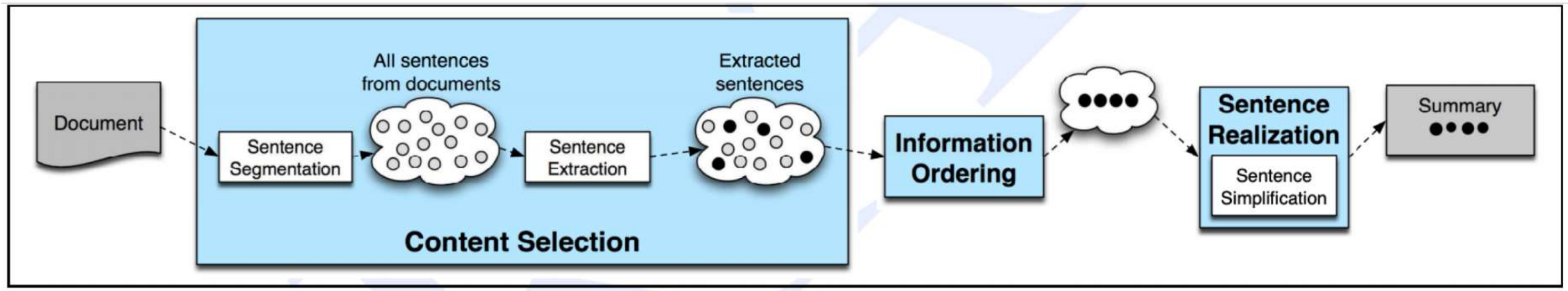


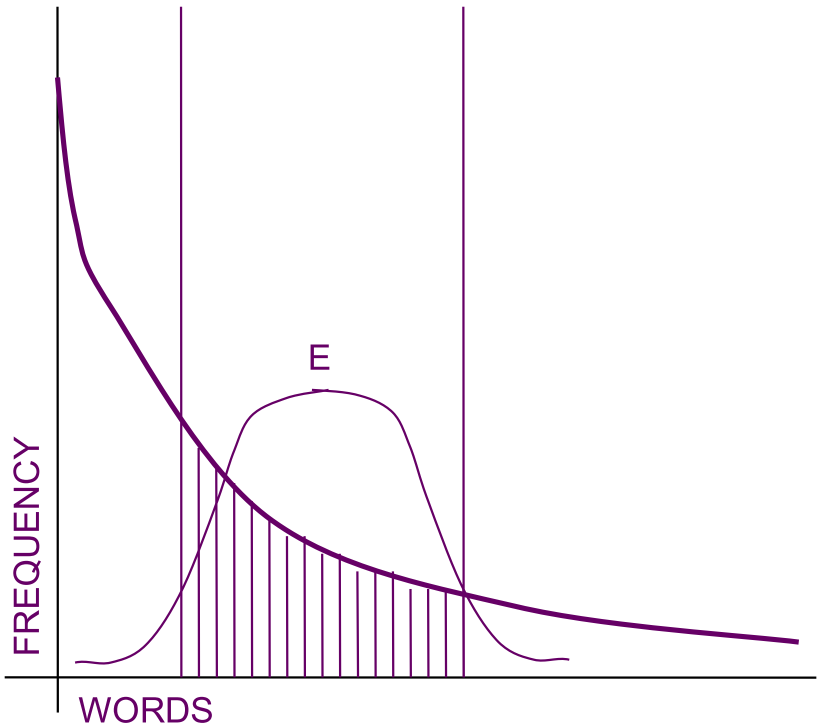
Figure 23.14 The basic architecture of a generic single document summarizer.

- Statistical / IR based Approach
  - Operate at lexical level, use word frequencies, similarity measures, etc.
  - Does not support abstraction.
- NLP / IE based Approach
  - Try to “understand” text. Needs rules for text analysis and manipulation.
  - Higher quality, supports abstraction.

Statistical / IR based approaches

Exploiting Word-frequency Information

- High frequency words are related to the topic of the document
  - Of course, this does not include stopwords
- Importance of sentence depends of
  - Number of occurrences of significant words
  - Discriminating power of the words
- Rank sentences and pick the top k



Resolving power of significant words

Exploiting Document Structure

- Some words/phrases positively correlated to summary

- Information from Structure
  - Title words
  - Section, sub-section heading words

- Information from Position
  - Genre dependent
  - First sentence of document, first sentence of paragraph, last sentence of document, etc.

Graph Based Methods

- **Key Idea**: Summarizing sentences are well connected to other sentences

- Connectivity based on similarity with other sentences

- Similarity measure: tf-idf could be used

- Graph G (V,E)
  - V: set of sentences
  - E: similarity between sentences > threshold

Graph Based Methods

Degree of Centrality

- **Key Idea**: Summarizing sentences are well connected to other sentences

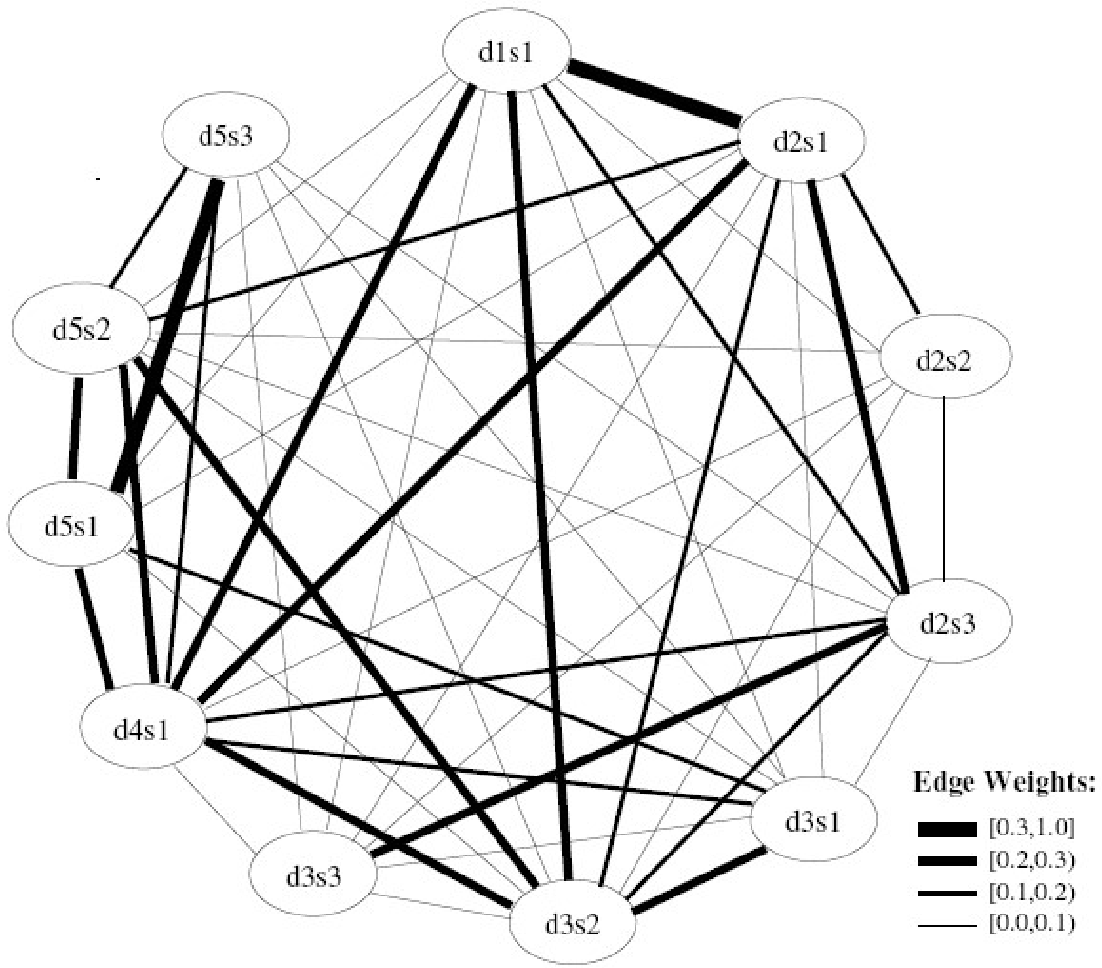
- Rank sentences by their degree

- Pick top k as summarizing sentences

- Sensitive to distortion by 'rogue' sentences

Graph Based Methods

Sentence Clusters based on Similarity



LexRank

- Inspired by PageRank
- Value connections from highly connected neighbours
- Random Markov Walk over the graph
- LR(u)= ∑ LR(v) / deg(v), where v is a neighbour of u

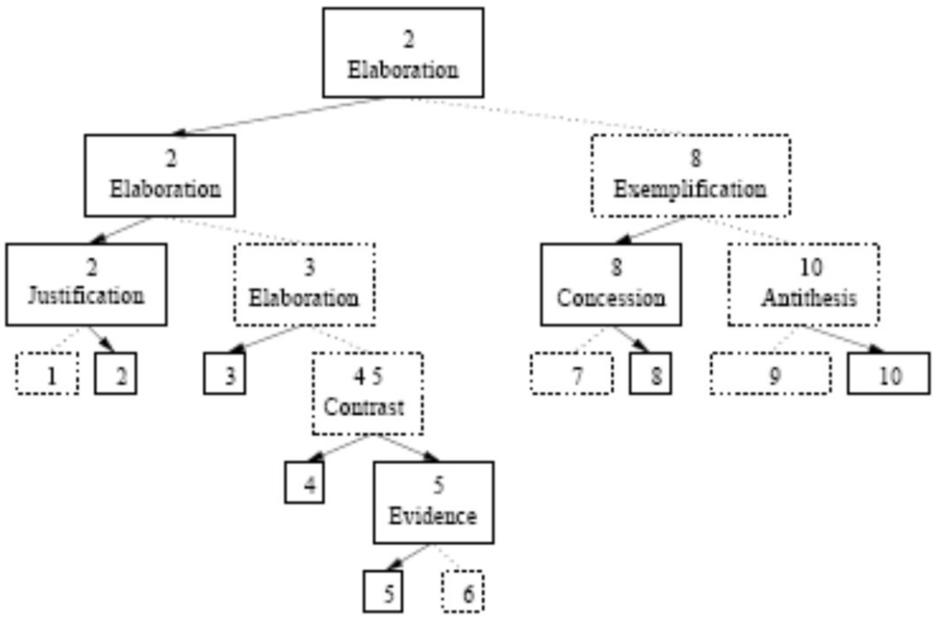
NLP / IE based Approaches

Rhetoric based Summarization

- Rhetoric Relation
  - Between two non overlapping spans of text
  - Nucleus : core idea
  - Satellite : arguments to favor core idea

- Rhetoric relation is a relation between Nucleus and Satellite.
  E.g. Justification, elaboration, contrast, evidence, etc.

Rhetoric based Summarization



Rhetoric Structure Tree

[With its distant orbit — 50 percent farther from the sun than Earth — and slim atmospheric blanket,[1]] [Mars experiences frigid weather conditions.[2]] [Surface temperatures typically average about −60 degrees Celsius (−76 degrees Fahrenheit) at the equator and can dip to −123 degrees C near the poles.[3]] [Only the midday sun at tropical latitudes is warm enough to thaw ice on occasion,[4]] [but any liquid water formed in this way would evaporate almost instantly[5]] [because of the low atmospheric pressure.[6]]

[Although the atmosphere holds a small amount of water, and water-ice clouds sometimes develop,[7]] [most Martian weather involves blowing dust or carbon dioxide.[8]] [Each winter, for example, a blizzard of frozen carbon dioxide rages over one pole, and a few meters of this dry-ice snow accumulate as previously frozen carbon dioxide evaporates from the opposite polar cap.[9]] [Yet even on the summer pole, where the sun remains in the sky all day long, temperatures never warm enough to melt frozen water.[10]]

Rhetoric based Summarization

Summarization Method

- Generate Rhetoric Structure Tree
  - Because of rhetoric ambiguity there are multiple trees
- Pick best tree using
  - Clustering-based metric
  - Shape-based metric etc.
- Pick up top K nodes nearest to the root, where K is number of sentences expected in summary

Summarization evaluation: ROUGE

ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

$$\text{ROUGE-N} = \frac{\sum_{S \in \{ReferemceSummaries\}} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in \{ReferenceSummaries\}} \sum_{gram_n \in S} Count(gram_n)} \quad (1)$$

Like BLEU, it's based on n-gram overlap. Differences:

- ROUGE has no brevity penalty
- ROUGE is based on recall, while BLEU is based on precision
    - Arguably, precision is more important for MT (then add brevity penalty to fix under-translation), and recall is more important for summarization (assuming you have a max length constraint)
    - However, often a F1 (combination of precision and recall) version of ROUGE is reported anyway.

Summarization evaluation: ROUGE

- BLEU is reported as a single number, which is combination of the precisions for n=1,2,3,4 n-grams

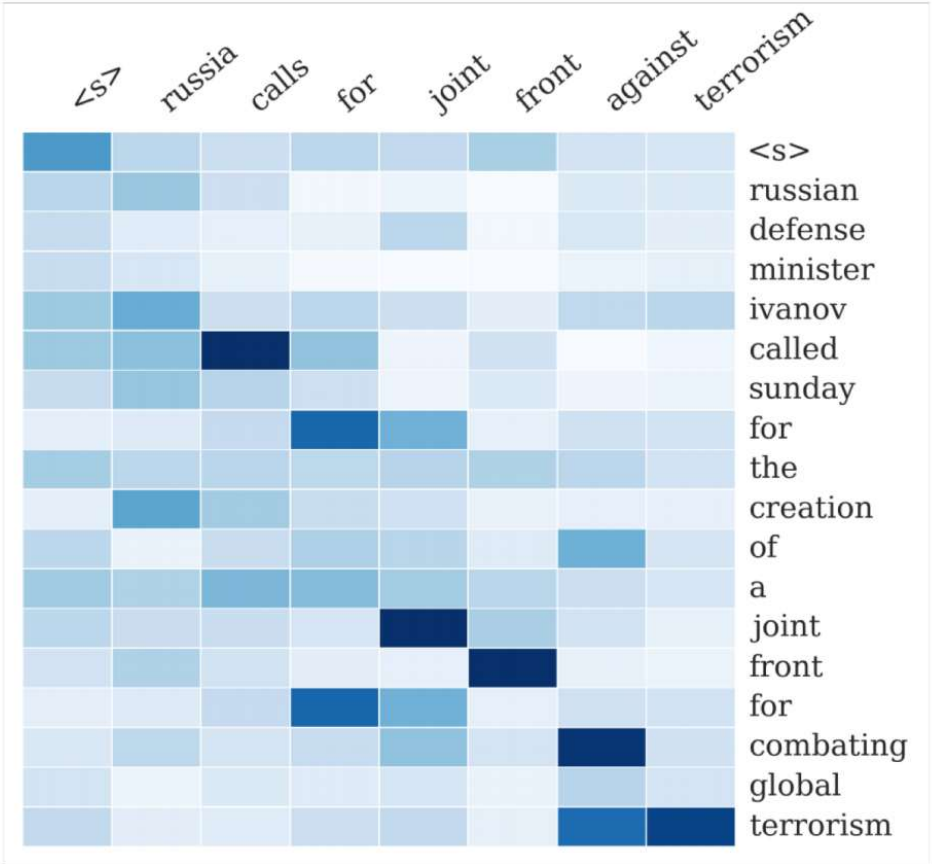
- ROUGE scores are reported separately for each n-gram

- The most commonly-reported ROUGE scores are:
  - ROUGE-1:* unigram overlap
  - ROUGE-2: bigram overlap
  - ROUGE-L: longest common subsequence overlap

Neural Approaches

Neural summarization (2015 - 2022)

- 2015: Rush et al. publish the first seq2seq summarization paper
- Single-document abstractive summarization is a translation task!
- Thus, we can apply standard seq2seq + attention NMT methods



A Neural Attention Model for Abstractive Sentence Summarization, Rush et al, 2015 [*https://arxiv.org/pdf/1509.00685.pdf*](https://arxiv.org/pdf/1509.00685.pdf)

Neural summarization (2015 - 2022)

Since 2015, there have been lots more developments!

- Making it easier to copy, but also preventing too much copying!
- Hierarchical / multi-level attention
- More global / high-level content selection
- Using Reinforcement Learning to directly maximize ROUGE, or other discrete goals (e.g. length)
- Resurrecting pre-neural ideas (e.g. graph algorithms for content selection) and working them into neural systems
- ….

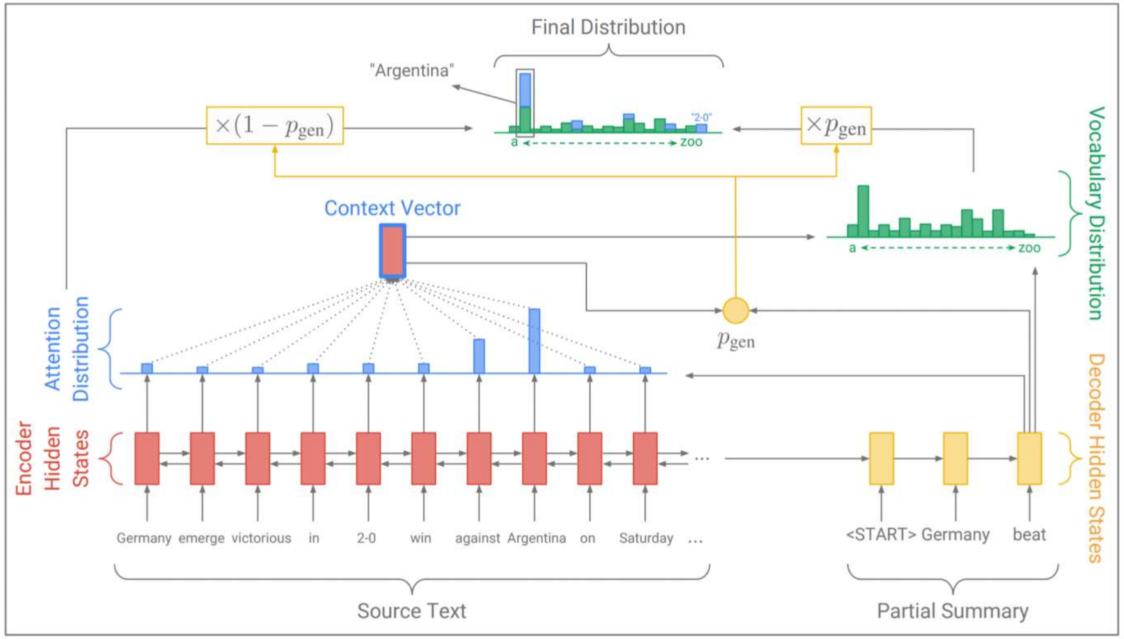
Copy mechanisms

- Seq2seq+attention systems are good at writing fluent output, but bad at copying over details (like rare words) correctly

- Copy mechanisms use attention to enable a seq2seq system to easily copy words and phrases from the input to the output
  - Clearly this is very useful for summarization
  - Allowing both copying and generating gives us a hybrid extractive/abstractive approach

- There are several papers proposing copy mechanism variants:
  - Language as a Latent Variable: Discrete Generative Models for Sentence Compression, Miao et al, 2016 https://arxiv.org/pdf/1609.07317.pdf
  - Abstractive Text Summarization using Sequence-to-sequence RNNs and Beyond, Nallapati et al, 2016 https://arxiv.org/pdf/1602.06023.pdf
  - Incorporating Copying Mechanism in Sequence-to-Sequence Learning, Gu et al, 2016 https://arxiv.org/pdf/1603.06393.pdf
  - etc

Copy mechanisms



<u>One example of how to do a copying mechanism</u>:

On each decoder step, calculate p_{gen}, the probability of generating the next word (rather than copying it). The final distribution is a mixture of the generation (aka "vocabulary") distribution, and the copying (i.e. attention) distribution:

$$\boxed{P(w) = p_{\text{gen}} P_{\text{vocab}}(w) + (1 - p_{\text{gen}}) \sum_{i:w_i=w} a_i^t}$$

Get To The Point: Summarization with Pointer-Generator Networks, See et al, 2017 https://arxiv.org/pdf/1704.04368.pdf

Copy mechanisms

Big problem with copying mechanisms:

- **They copy too much!**
  - Mostly long phrases, sometimes even whole sentences
- What should be an abstractive system collapses to a mostly extractive system.

Another problem:

- They're bad at overall content selection, especially if the input document is long
- No overall strategy for selecting content

Better content selection

- Recall: pre-neural summarization had separate stages for **content selection** and **surface realization** (i.e. text generation)

- In a standard seq2seq+attention summarization system, these two stages are mixed in together
  - On each step of the decoder (i.e. surface realization), we do word-level content selection (attention)
  - This is bad: no global content selection strategy

- One solution: bottom-up summarization

Bottom-up summarization

- Content selection stage: Use a neural sequence-tagging model to tag words as include or don’t-include
- Bottom-up attention stage: The seq2seq+attention system can’t attend to words tagged don’t-include (apply a mask)

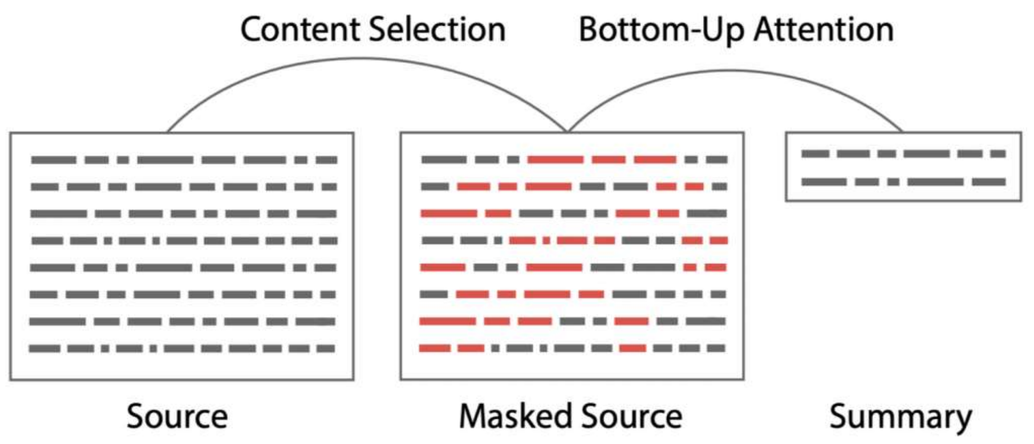


Figure 2: Overview of the selection and generation processes described throughout Section 4.

Simple but effective!

- Better overall content selection strategy
- Less copying of long sequences (i.e. more abstractive output)

Bottom-Up Abstractive Summarization, Gehrmann et al, 2018 https://arxiv.org/pdf/1808.10792v1.pdf

Reinforcement Learning (RL)

- In 2017 Paulus et al published a "deep reinforced" summarization model
- Main idea: Use Reinforcement Learning (RL) to directly optimize ROUGE-L
    - By contrast, standard maximum likelihood (ML) training can't directly optimize ROUGE-L because it's a non-differentiable function
- Interesting finding:
    - Using RL instead of ML achieved higher ROUGE scores, but lower human judgment scores

Model	ROUGE-1	ROUGE-2	ROUGE-L
ML, no intra-attention	44.26	27.43	40.41
ML, with intra-attention	43.86	27.10	40.11
RL, no intra-attention	**47.22**	30.51	**43.27**
ML+RL, no intra-attention	47.03	**30.72**	43.10

Model	Readability	Relevance
ML	6.76	7.14
RL	4.18	6.32
ML+RL	**7.04**	**7.45**

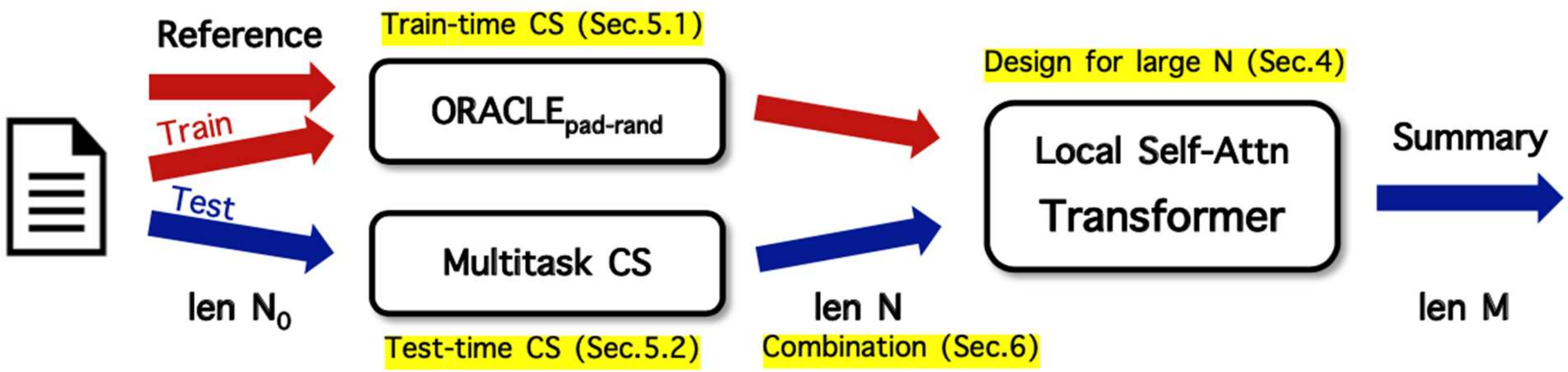
"We observed that our models with the highest ROUGE scores also generated barely-readable summaries."

Overall, a hybrid approach does best!

A Deep Reinforced Model for Abstractive Summarization, Paulus et al, 2017 https://arxiv.org/pdf/1705.04304.pdf

Long-span summarization

- In 2021 Manakul et al published a summarization model for long document summarization
- Constraining attention mechanism to be local, allowing longer input spans during training
- To reduce memory and compute requirements, perform content selection explicitly before the abstractive stage
    - At training time, they investigate methods to select data for training fixed-span abstractive models
    - At test time, they ranks sentences through extractive labelling-based module and attention-based module



Long-Span Summarization via Local Attention and Content Selection https://aclanthology.org/2021.acl-long.470.pdf

Cross-Lingual Summarization

Dialogue
Blair: Remember we are seeing the wedding planner after work
Chuck: Sure, where are we meeting her?
Blair: At Nonna Rita's
Chuck: Can I order their seafood tagliatelle or are we just having coffee with her? I've been dreaming about it since we went there last month
Blair: Haha sure why not
Chuck: Well we both remmber the spaghetti pomodoro disaster from our last meeting with Diane
Blair: Omg hahaha it was all over her white blouse
Chuck: :D
Blair: :P
Summary
Blair and Chuck are going to meet the wedding planner after work at Nonna Rita's. The tagliatelle served at Nonna Rita's are very good.


```
{
  "id": "NPR-11",
  "program": "Day to Day",
  "date": "2008-06-10",
  "url": "https://www.npr.org/templates/story/story.php?storyId=91356794",
  "title": "Researchers Find Discriminating Plants",
  "summary": "The 'sea rocket' shows preferential treatment to plants that are its kin. Evolutionary
  plant ecologist Susan Dudley of McMaster University in Ontario discusses her discovery.",
  "utt": [
    "This is Day to Day. I'm Madeleine Brand.",
    "And I'm Alex Cohen.",
    "Coming up, the question of who wrote a famous religious poem turns into a very unchristian
    battle.",
    "First, remember the 1970s? People talked to their houseplants, played them classical music.
    They were convinced plants were sensuous beings and there was that 1979 movie, 'The Secret
    Life of Plants.'",
    "Only a few daring individuals, from the scientific establishment, have come forward with
    offers to replicate his experiments, or test his results. The great majority are content simply
    to condemn his efforts without taking the trouble to investigate their validity.",
    ...
    "OK. Thank you.",
    "That's Susan Dudley. She's an associate professor of biology at McMaster University in
    Hamilt on Ontario. She discovered that there is a social life of plants."
  ],
  "speaker": [
    "MADELEINE BRAND, host",
    "ALEX COHEN, host",
    "ALEX COHEN, host",
    "MADELEINE BRAND, host",
    "Unidentified Male",
    ...
    "Professor SUSAN DUDLEY (Biology, McMaster University)",
    "MADELEINE BRAND, host"
  ]
}
```


Wang, J., Meng, F., Lu, Z., Zheng, D., Li, Z., Qu, J., & Zhou, J. (2022). Clidsum: A benchmark dataset for cross-lingual dialogue summarization. EMNLP 2022.

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Unsupervised Summarization

Chat Log

A: Is anyone there, please?

A: I want to buy this skirt, but I don't know what size suits me.
B: What's your height and weight?
A: 165cm and 55kg.
B: Well, size M suits you.

A: How much? The store page says that coupons can be claimed.
B: 588 yuan. A coupon of 20 yuan is available for orders over 500.
A: Can I use it during shopping festival on November 11th?
B: Sorry. Not supported.

A: Okey. Which courier company?
B: We all use SF Express.
A: Free shipping?
B: Yeah. Free shipping is offered if you spend at least 300 yuan.
A: I have placed an order. When will it be delivered?
B: Tomorrow.

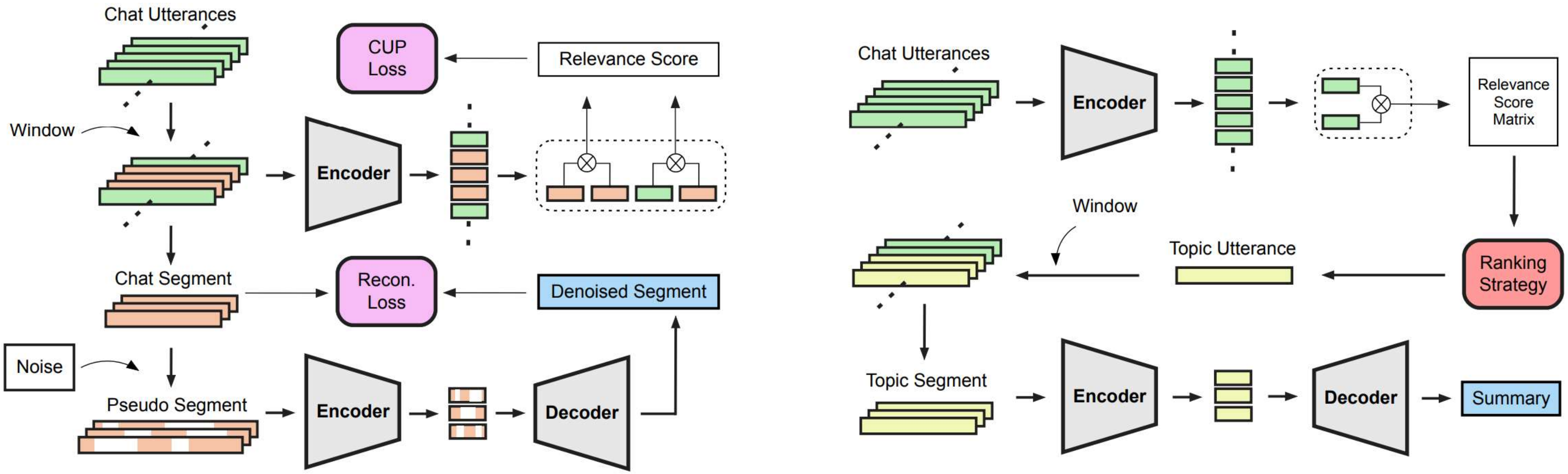
Summary

(Topic 1) The user wants to buy a skirt. Size M suits people of 165cm and 55kg. **(Topic 2)** The skirt costs 588 yuan. A coupon of 20 yuan is available, which is not supported during November 11th. **(Topic 3)** The skirt will be delivered tomorrow by SF Express with free shipping.

Example of a chat log and its reference summary. It has three topics: **Skirt Size**, **Price** and **Logistics**. Utterances in the same color box describe the same topic. An interrogative sentence with its elliptical response is underlined.

Zou, Y., Lin, J., Zhao, L., Kang, Y., Jiang, Z., Sun, C., ... & Liu, X. (2021, May). Unsupervised summarization for chat logs with topic-oriented ranking and context-aware auto-encoders. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 16, pp. 14674-14682).

Unsupervised Summarization



(a) The training stage of RankAE.

(b) The inference stage of RankAE.

Zou, Y., Lin, J., Zhao, L., Kang, Y., Jiang, Z., Sun, C., ... & Liu, X. (2021, May). Unsupervised summarization for chat logs with topic-oriented ranking and context-aware auto-encoders. In Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 16, pp. 14674-14682).

Dialogue

Dialogue

"Dialogue" encompasses a large variety of settings:

- **Task-oriented dialogue**
  - Assistive (e.g. customer service, giving recommendations, question answering, helping user accomplish a task like buying or booking something)
  - Co-operative (two agents solve a task together through dialogue)
  - Adversarial (two agents compete in a task through dialogue)
- **Social dialogue**
  - Chit-chat (for fun or company)
  - Therapy / mental wellbeing

Pre- and post-neural dialogue

- Due to the difficulty of open-ended freeform NLG, pre-neural dialogue systems more often used predefined templates, or retrieve an appropriate response from a corpus of responses.

- As in summarization research, since 2015 there have been many papers applying seq2seq methods to dialogue – thus leading to a renewed interest in open-ended freeform dialogue systems

- Some early seq2seq dialogue papers include:
  - *A Neural Conversational Model, Vinyals et al, 2015*
    https://arxiv.org/pdf/1506.05869.pdf
  - *Neural Responding Machine for Short-Text Conversation, Shang et al, 2015*
    https://www.aclweb.org/anthology/P15-1152

Seq2seq-based dialogue

However, it quickly became apparent that a naive application of standard seq2seq+attention methods has serious pervasive deficiencies for (chitchat) dialogue:

- Generic / boring responses
- Irrelevant responses (not sufficiently related to context)
- Repetition
- Lack of context (not remembering conversation history)
- Lack of consistent persona

Irrelevant response problem

Problem: seq2seq often generates response that's unrelated to user's utterance

- Either because it's generic (e.g. "I don't know")
- Or because changing the subject to something unrelated

One solution: optimize for Maximum Mutual Information (MMI) between input S and response T:

$$\log \frac{p(S,T)}{p(S)p(T)}$$

$$\hat{T} = \arg\max_{T} \left\{ \log p(T|S) - \log p(T) \right\}$$

A Diversity-Promoting Objective Function for Neural Conversation Models, Li et al, 2016 https://arxiv.org/pdf/1510.03055.pdf

Generic / boring response problem

Easy test-time fixes:

- Directly upweight rare words during beam search
- Use a sampling decoding algorithm rather than beam search

Conditioning fixes:

- Condition the decoder on some additional content (e.g. sample some content words and attend to them)
- Train a retrieve-and-refine model rather than a generate-from-scratch model
  - i.e. sample an utterance from your corpus of human-written utterances, and edit it to fit the current scenario.
  - This usually produces much more diverse / human-like / interesting utterances!

Why are Sequence-to-Sequence Models So Dull?, Jiang et al, 2018
https://staff.fnwi.uva.nl/m.derijke/wp-content/papercite-data/pdf/jiang-why-2018.pdf

Repetition problem

Simple solution:

- Directly block repeating n-grams during beam search.
  - Usually pretty effective!

More complex solutions:

- Train a coverage mechanism – in seq2seq, this is an objective that prevents the attention mechanism from attending to the same words multiple times.
- Define a training objective to discourage repetition
  - If this is a non-differentiable function of the generated output, then will need some technique like e.g. RL to train

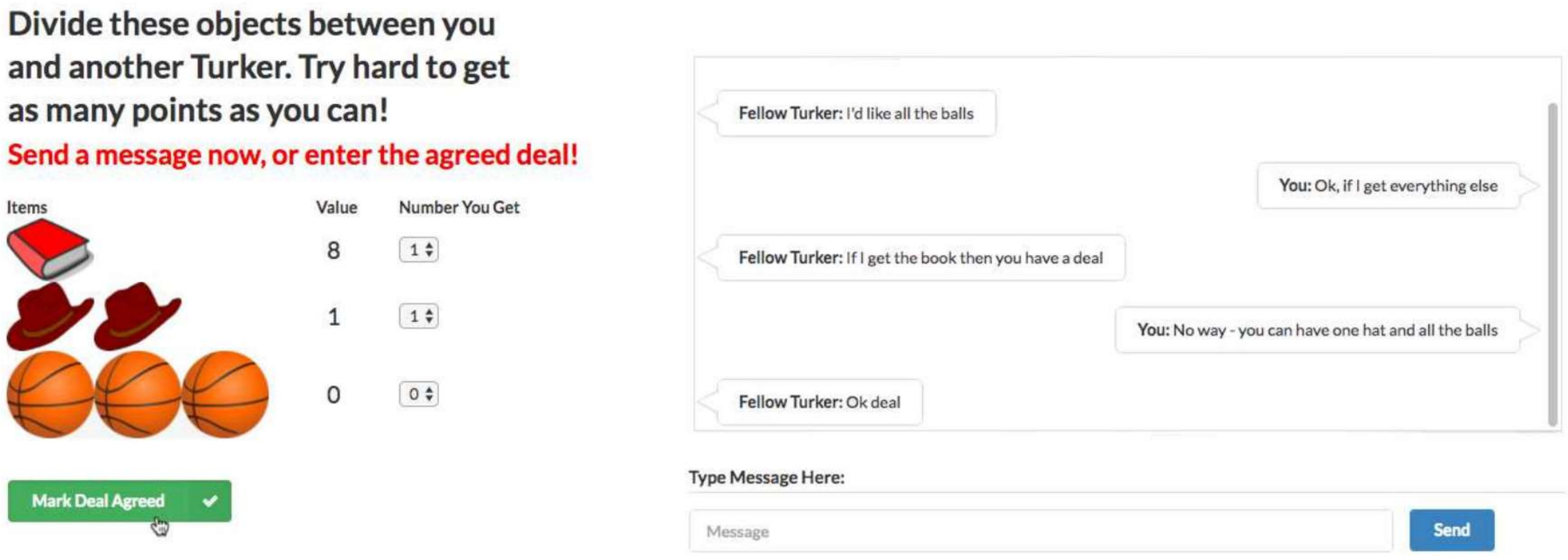
Lack of consistent persona problem

- In 2016, Li et al proposed a seq2seq dialogue model that learns to encode both conversation partners' personas as embeddings
  - The generated utterances are conditioned on the embeddings
- Similarly, a chitchat dataset called PersonaChat, which includes personas (collections of 5 sentences describing personal traits) for every conversation.
  - This provides a light type of grounding, allowing researchers to build persona-conditional dialogue agents
- Recently, Wang et al utilized tensor factorization to model users' personalization information with their posting histories
  - The personalized response embedding is fed to decoder to help generate more personalized responses.

A Persona-Based Neural Conversation Model, Li et al 2016, https://arxiv.org/pdf/1603.06155.pdf
Personalizing Dialogue Agents: I have a dog, do you have pets too?, Zhang et al, 2018 https://arxiv.org/pdf/1801.07243.pdf
Personalized Response Generation with Tensor Factorization, Wang et al, 2021 https://aclanthology.org/2021.gem-1.5.pdf

Negotiation dialogue

- In 2017, Lewis et al collected a negotiation dialogue dataset
- Two agents negotiate (via natural language) how to divide a set of items.
- The agents have different value functions for the items.
- The agents talk until they reach an agreement.



Deal or No Deal? End-to-End Learning for Negotiation Dialogues, Lewis et al, 2017 https://arxiv.org/pdf/1706.05125.pdf

Negotiation dialogue

- They find that training seq2seq systems for the standard maximum likelihood (ML) objective yields fluent but strategically poor dialogue agents.
- Like the Paulus et al summarization paper, they use Reinforcement Learning to optimize for a discrete reward (with the agents playing against themselves during training)
- The RL goal-based objective is combined with the ML objective
- Potential pitfall: If the agents just optimize just the RL goal while playing against each other, they might diverge from English*

Deal or No Deal? End-to-End Learning for Negotiation Dialogues, Lewis et al, 2017 https://arxiv.org/pdf/1706.05125.pdf

Negotiation dialogue

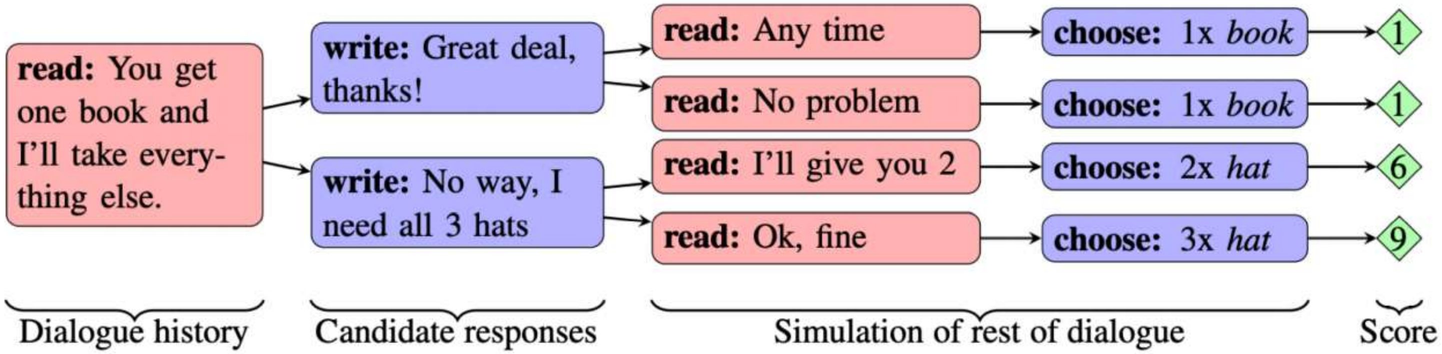


Figure 4: Decoding through rollouts: The model first generates a small set of candidate responses. For each candidate it simulates the future conversation by sampling, and estimates the expected future reward by averaging the scores. The system outputs the candidate with the highest expected reward.

- At test time, the model chooses between possible responses by computing rollouts: simulations of the rest of the conversation and the expected reward.

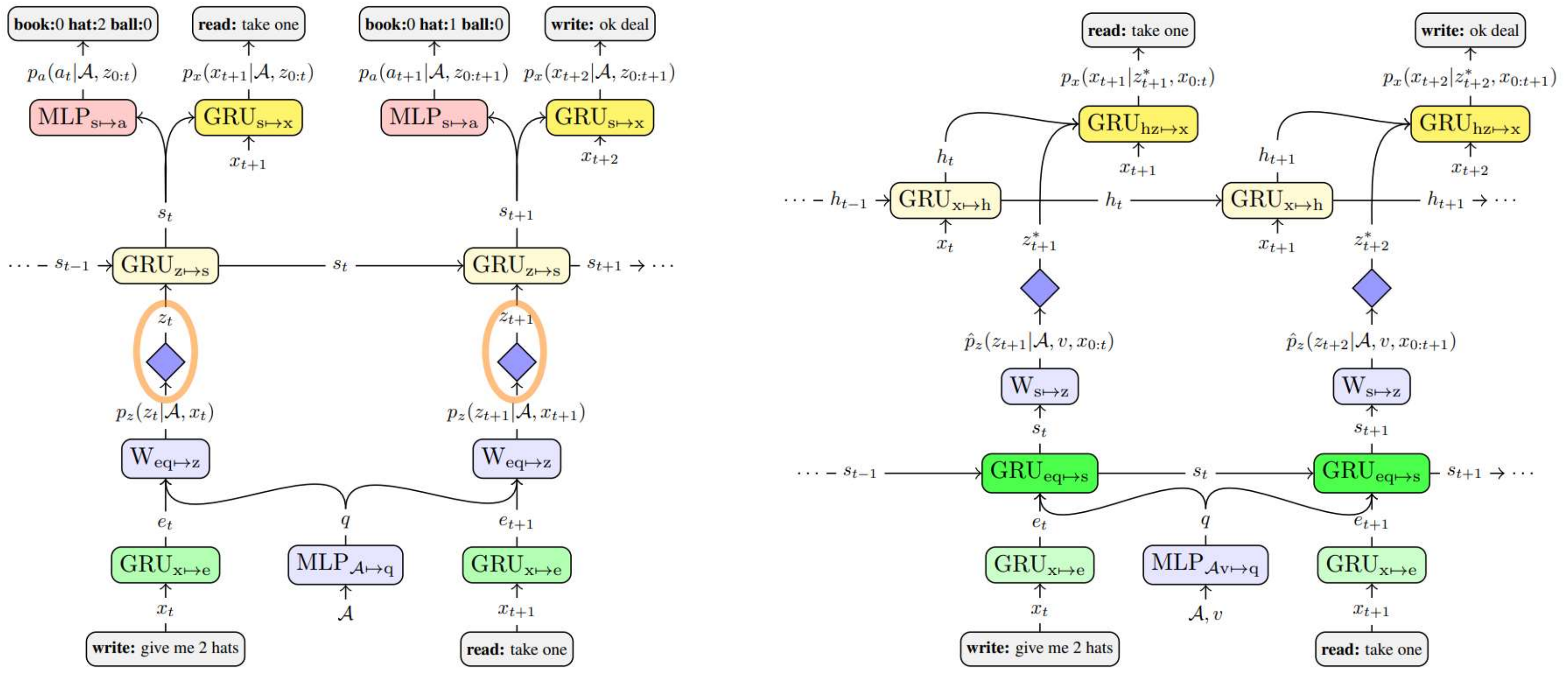
Deal or No Deal? End-to-End Learning for Negotiation Dialogues, Lewis et al, 2017 https://arxiv.org/pdf/1706.05125.pdf

Negotiation dialogue

- In 2018, Yarats et al proposed another dialogue model for the negotiation task, that separates the strategic aspect from the NLG aspect.
- Each utterance x_t has a corresponding discrete latent variable z_t
- z_t is learnt to be a good predictor of future events in the dialogue (future messages, ultimate strategic outcome), but not a predictor of xt itself
- This means that "z_t learns to represent x_t 's effect on the dialogue, but not the words of x_t "
- Thus z_t separates the strategic aspect of the task from the NLG aspect. This is useful for controllability, interpretability, easier to learn strategy, etc.

Hierarchical Text Generation and Planning for Strategic Dialogue, Yarats et al, 2018 https://arxiv.org/pdf/1712.05846.pdf

Negotiation dialogue



(a) Clustering Model

(b) Full Model

Figure 3: We pre-train a model to learn a discrete encoder for sentences, which bottlenecks the message x_t through discrete representation z_t (Figure 3a; §5). This architecture forces z_t to capture the most relevant aspects of x_t for predicting future messages and actions. We then extract the learned discrete representations z_t (marked by orange ellipses) and train our full model (Figure 3b): p_x is trained to translate representations z_t^* into messages x_t (§6.1), and $\hat{p}_z$ is trained to predict a distribution over z_t given the dialogue history (§6.2).

Hierarchical Text Generation and Planning for Strategic Dialogue, Yarats et al, 2018 https://arxiv.org/pdf/1712.05846.pdf

Storytelling

Storytelling

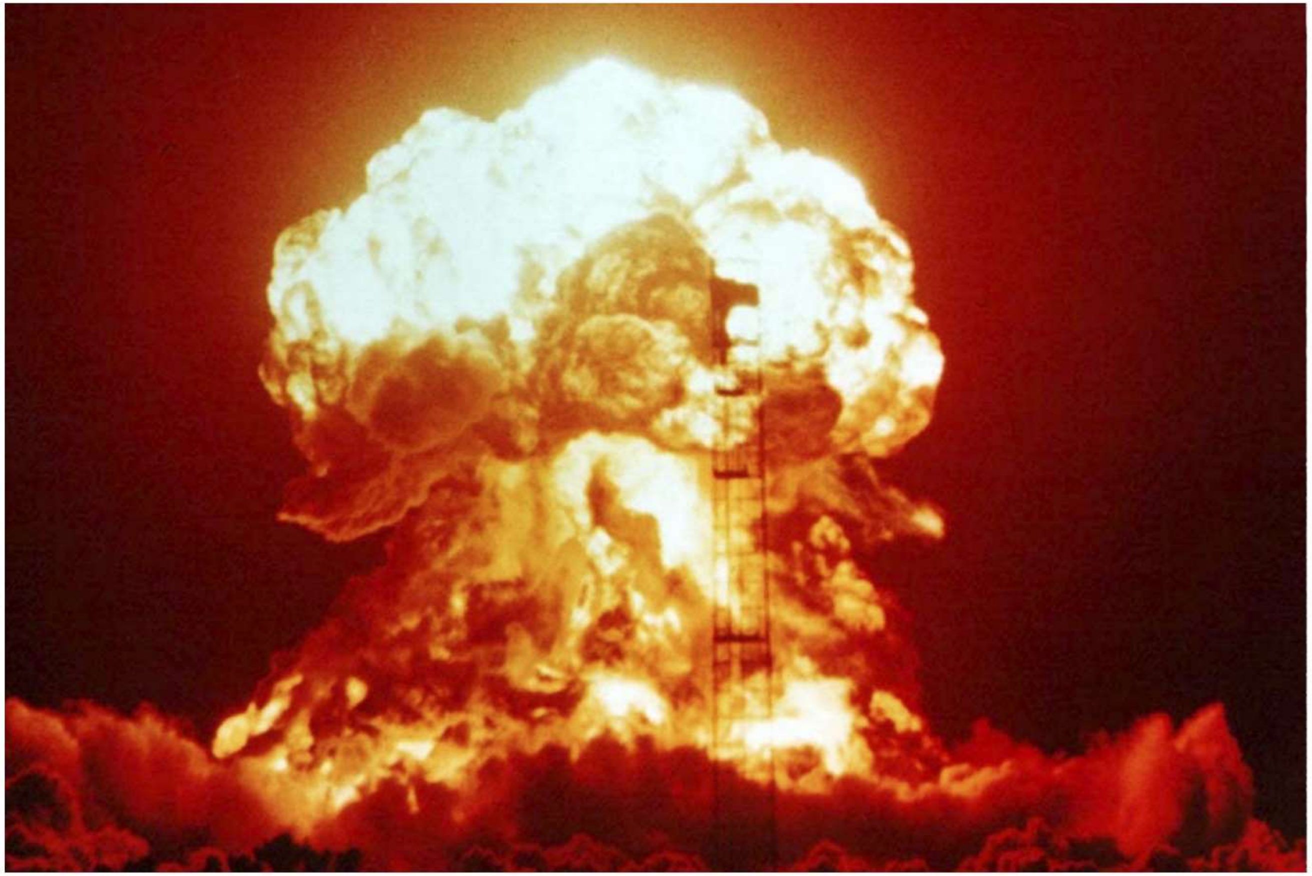
Most neural storytelling work uses some kind of prompt

- Generate a story-like paragraph given an image
- Generate a story given a brief writing prompt
- Generate the next sentence of a story, given the story so far (story continuation)
  - This is different to the previous two, because we are not concerned with the system's performance over several generated sentences

Neural storytelling is taking off!

- The first Storytelling Workshop was held in 2018
- It held a competition (generate a story to accompany a sequence of 5 images)

Generating a story from an image



Generated story about image
Model: Taylor Swift Lyrics

"You have to be the only light bulb in the night sky, I thought, Oh, God, it 's so dark out of me that I missed you , I promise."

What's interesting here is that this isn't straightforward supervised image-captioning. There was no paired data to learn from.

Generating a story from an image

- Question: How to get around the lack of parallel data?
- Answer: Use a common sentence-encoding space

- Skip-thought vectors are a type of general-purpose sentence embedding method
  - The idea is similar to how we learn an embedding for a word by trying to predict the words around it
- Using COCO (an image captioning dataset), learn a mapping from images to the skip-thought encodings of their captions
- Using the target style corpus (Taylor Swift lyrics), train a RNN-LM to decode a skip-thought vector to the original text
- Put the two together

Generating a story from a writing prompt

- In 2018, Fan et al released a new story generation dataset collected from Reddit's WritingPrompts subreddit.
- Each story has an associated brief writing prompt.

Prompt: The Mage, the Warrior, and the Priest

Story: A light breeze swept the ground, and carried with it still the distant scents of dust and time-worn stone. The Warrior led the way, heaving her mass of armour and muscle over the uneven terrain. She soon crested the last of the low embankments, which still bore the unmistakable fingerprints of haste and fear. She lifted herself up onto the top the rise, and looked out at the scene before her. [...]

Hierarchical Neural Story Generation, Fan et al, 2018 https://www.aclweb.org/anthology/P18-1082.pdf

Generating a story from a writing prompt

Fan et al also proposed a complex seq2seq prompt-to-story model:

- It's convolutional-based
  - This makes it faster than RNN-based seq2seq
- Gated multi-head multi-scale self-attention
  - The self-attention is important for capturing long-range context
  - The gates allow the attention mechanism to be more selective
  - The different attention heads attend at different scales – this means there are different attention mechanisms dedicated to retrieving finegrained information and coarse-grained information
- Model fusion:
  - Pretrain one seq2seq model, then train a second seq2seq model that has access to the hidden states of the first
  - The idea is that the first seq2seq model learns general LM and the second learns to condition on the prompt

Hierarchical Neural Story Generation, Fan et al, 2018 https://www.aclweb.org/anthology/P18-1082.pdf

Generating a story from a writing prompt

The results are impressive!

- Related to prompt
- Diverse; non-generic
- Stylistically dramatic

Example Prompt 1: Aliens start abducting humans

It has been two weeks, and the last of my kind has gone. It is only a matter of time until there will be nothing left. I'm not sure what the hell is going on... I can't think. I can hear a distant scream. I think of a strange, alien sound. I try to speak, but am interrupted by something, something that feels like a drum, I ca not tell. I mean I'm just a little bit older than an average human. But I can, and I can feel the vibrations . I hear the sound of a ship approaching. The ground quakes at the force of the impact, and a loud explosion shatters the silence.

However:

- Mostly atmospheric/descriptive/scene-setting; less events/plot
- When generating for longer, mostly stays on the same idea without moving forward to new ideas – coherence issues

Hierarchical Neural Story Generation, Fan et al, 2018 https://arxiv.org/pdf/1805.04833.pdf

Challenges in storytelling

Stories generated by neural LMs can sound fluent…
but are meandering, nonsensical, with no coherent plot

What's missing?
LMs model sequences of words. Stories are sequences of events.

To tell a story, we need to understand and model:

- Events and the causality structure between them
- Characters, their personalities, motivations, histories, and relationships to other characters
- State of the world (who and what is where and why)
- Narrative structure (e.g. exposition → conflict → resolution)
- Good storytelling principles (don't introduce a story element then never use it)

Event2event Story Generation

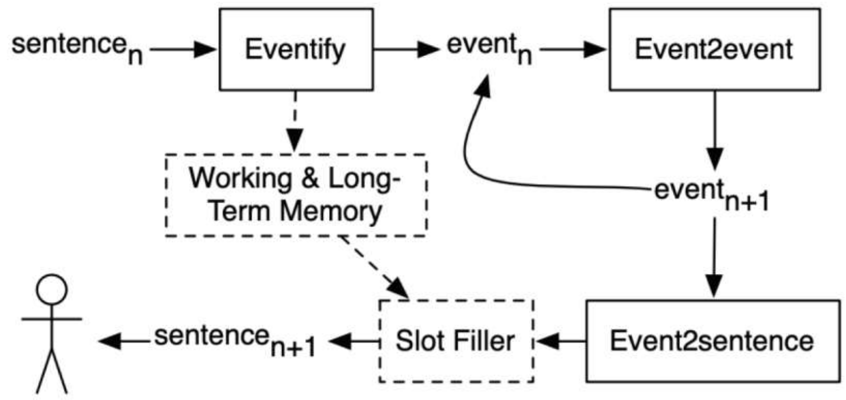


Figure 1: Our automated story generation pipeline. Dashed boxes and arrows represent future work.

Experiment	**Input**	**Extracted Event(s)**	**Generated Next Event(s)**	**Generated Next Sentence**
All Generalized Events & Generalized Sentence	He reaches out to Remus Lupin, a Defence Against the Dark Arts teacher who is eventually revealed to be a werewolf.	⟨male.n.02, get-13.5.1, ∅, <CHAR>0⟩ ⟨ORGANIZATION, say-37.7-1, monster.n.01, ∅⟩	⟨monster.n.01, amuse-31.1, sarge, ∅⟩ ⟨monster.n.01, amuse-31.1, realize, ∅⟩ ⟨monster.n.01, conjecture-29.5-1, ∅, ∅⟩ ⟨male.n.02, conduit.n.01, entity.n.01, ∅⟩ ⟨male.n.02, free-80-1, ∅, penal_institution.n.01⟩	When monster.n.01 nemesis.n.01 describes who finally realizes male.n.02 can not, dangerous entity.n.01 male.n.02 is released from penal_institution.n.01.

Event Representations for Automated Story Generation with Deep Neural Nets, Martin et al, 2018
https://www.aaai.org/ocs/index.php/AAAI/AAAI18/paper/view/17046/15769

Structured Story Generation

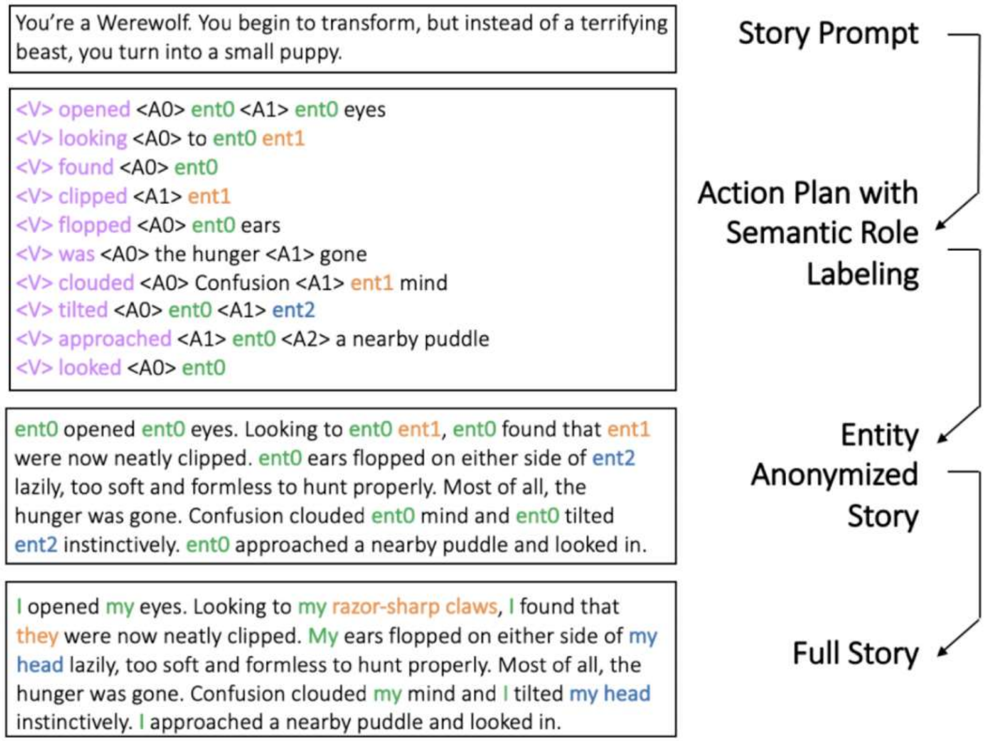


Figure 1: Proposed Model. Conditioned upon the prompt, we generate sequences of predicates and arguments. Then, a story is generated with placeholder entities such as *ent0*. Finally we replace the placeholders with specific references.

Strategies for Structuring Story Generation, Fan et al, 2019 https://www.aclweb.org/anthology/P19-1254.pdf

Long Stories Generation

Yang et al proposed TopNet model:

- Can alleviate the information sparsity problem
  - by leveraging the recent advances in neural topic modeling to obtain high-quality skeleton words to complement the short input
- Instead of directly generating a story
  - They first learn to map the short text input to a low-dimensional topic distribution
  - The distribution is pre-assigned by a topic model
- Based on this latent topic distribution
  - They can use the reconstruction decoder of the topic model to sample a sequence of inter-related words as a skeleton for the story
- Their proposed framework is highly effective in skeleton word selection and significantly outperforms the SOTA models

TopNet: Learning from Neural Topic Model to Generate Long Stories https/dl.acm.org/doi/pdf/10.1145/3447548.3467410

Long Stories Generation

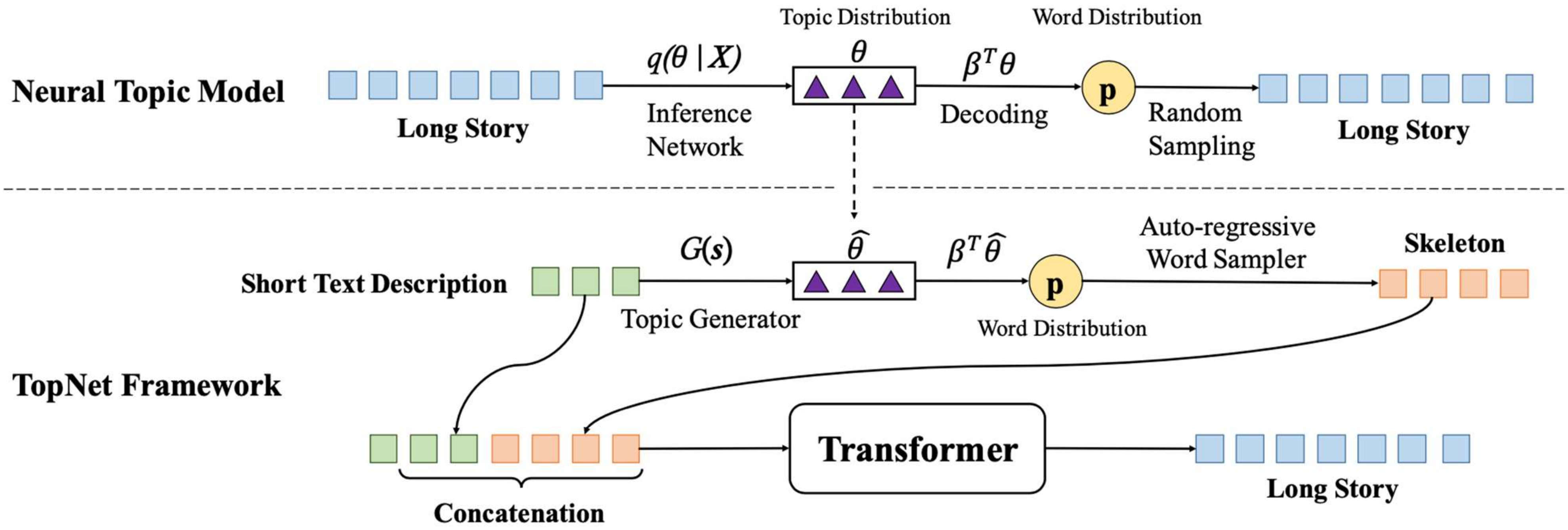
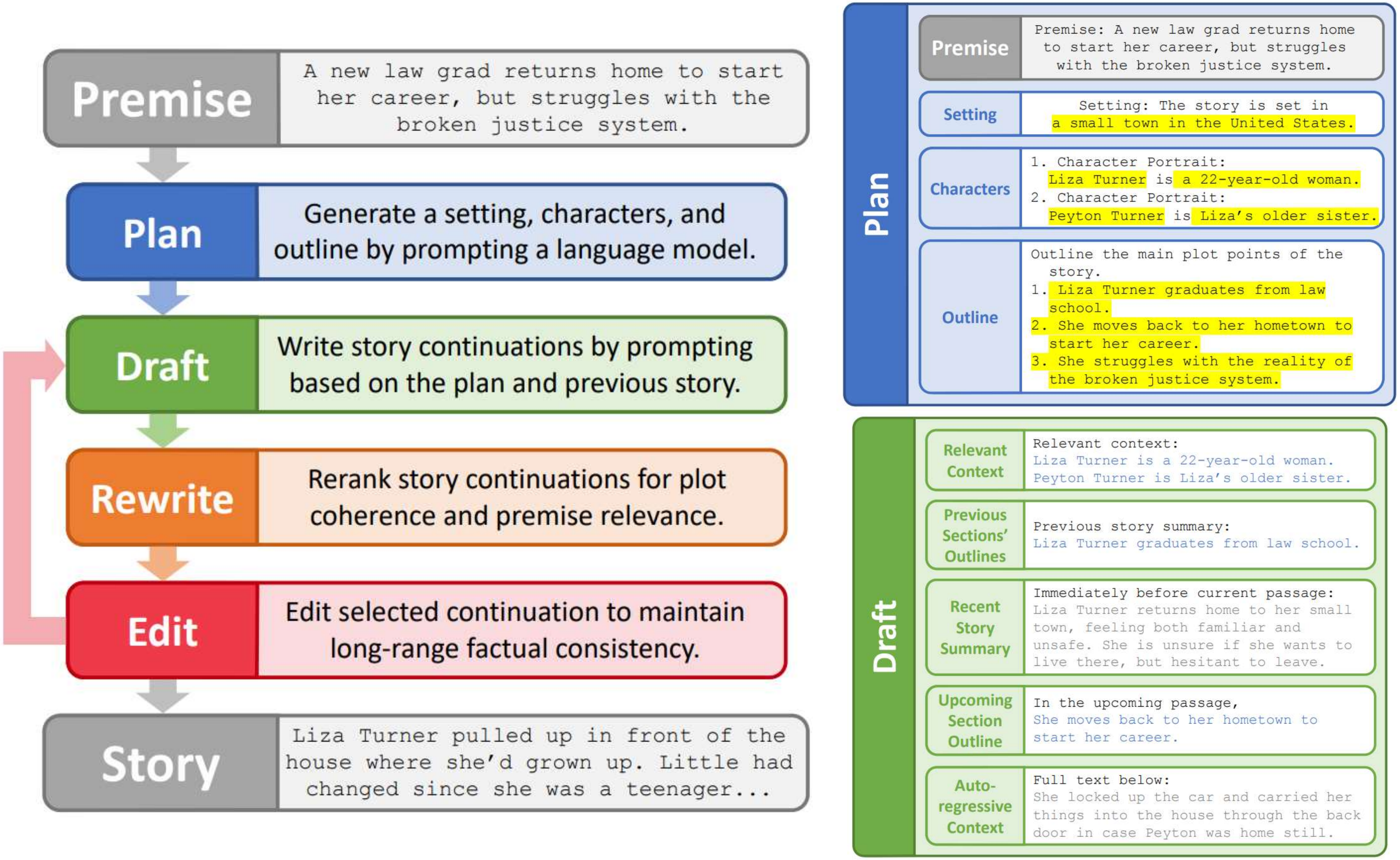


Figure 1: Overview of our model, comprising the neural topic model (upper) and TopNet framework (bottom).

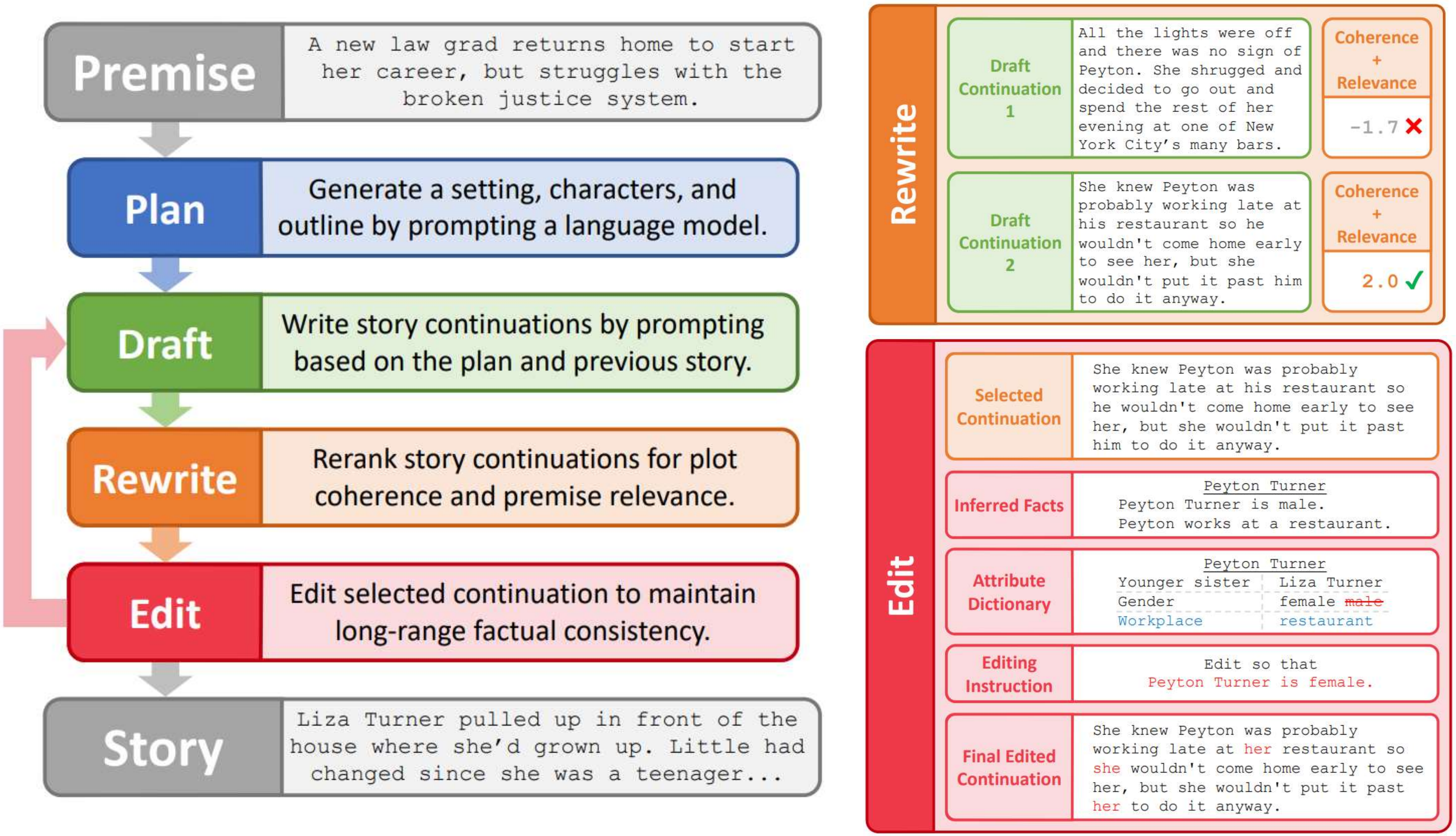
TopNet: Learning from Neural Topic Model to Generate Long Stories https/dl.acm.org/doi/pdf/10.1145/3447548.3467410

Long Stories Generation



Yang, K., Peng, N., Tian, Y., & Klein, D. (2022). Re3: Generating longer stories with recursive reprompting and revision. EMNLP 2022

Long Stories Generation



Yang, K., Peng, N., Tian, Y., & Klein, D. (2022). Re3: Generating longer stories with recursive reprompting and revision. EMNLP 2022

Poetry generation

Poetry generation: Hafez

- Hafez: a poetry generation system by Ghazvininejad et al
- Main idea: Use a Finite State Acceptor (FSA) to define all possible sequences that obey the desired rhythm constraints. Then use the FSA to constrain the output of a RNN-LM.

For example:

- A Shakespearean sonnet is 14 lines of iambic pentameter

one line of iambic pentameter →

```
   010      1   0     10       101
Attending on his golden pilgramage
```

- So the Shakespearean sonnet FSA is ((01)^5)^14
- During beam search decoding, only explore hypotheses that fall within the FSA.

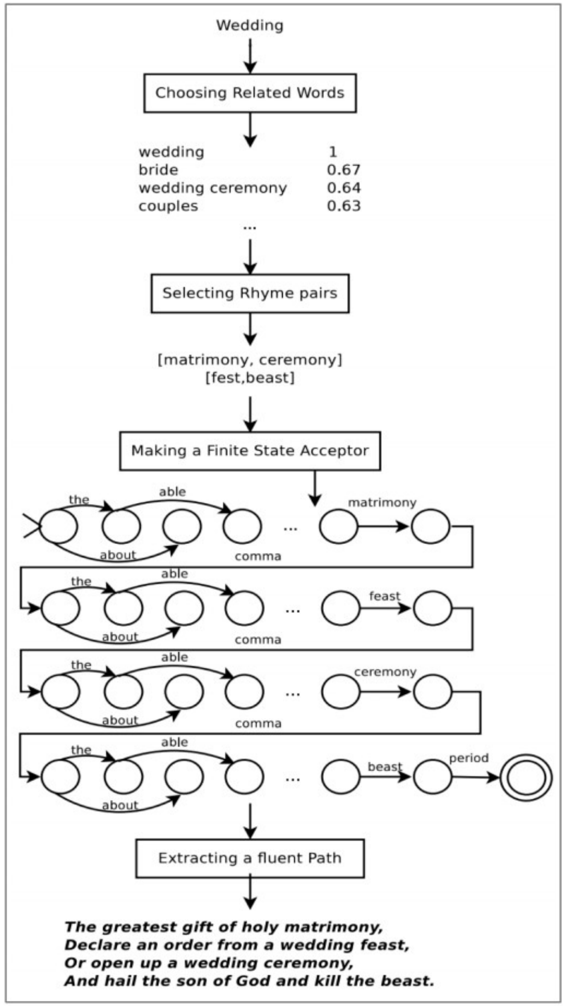
Generating Topical Poetry, Ghazvininejad et al, 2016 http://www.aclweb.org/anthology/D16-1126

Hafez: an Interactive Poetry Generation System, Ghazvininejad et al, 2017 http://www.aclweb.org/anthology/P17-4008

Poetry generation: Hafez

Full system:

- User provides topic word
- Get a set of words related to topic
- Identify rhyming topical words. These will be the ends of each line
- Generate the poem using RNN-LM constrained by FSA
- The RNN-LM is backwards (right-toleft). This is necessary because last word of each line is fixed.



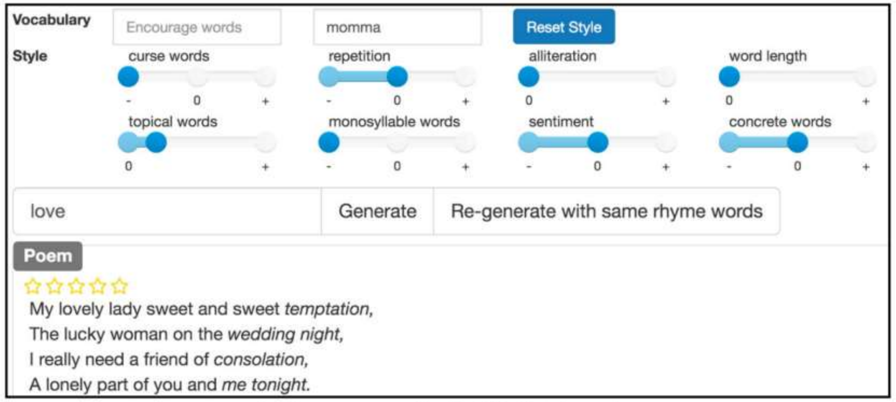
Generating Topical Poetry, Ghazvininejad et al, 2016 http://www.aclweb.org/anthology/D16-1126

Hafez: an Interactive Poetry Generation System, Ghazvininejad et al, 2017 http://www.aclweb.org/anthology/P17-4008

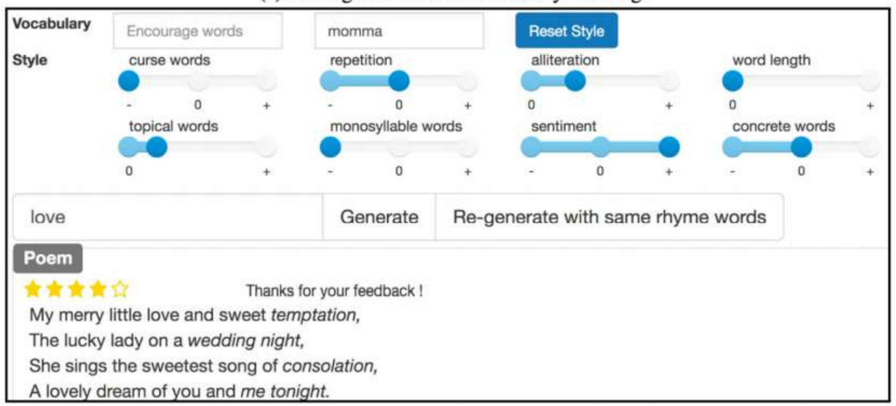
Poetry generation: Hafez

- In a follow-up paper, the authors made the system interactive and user-controllable.

- The control method is simple: during beam search, upweight the scores of words that have the desired features.



(a) Poem generated with default style settings



Generating Topical Poetry, Ghazvininejad et al, 2016 http://www.aclweb.org/anthology/D16-1126

Hafez: an Interactive Poetry Generation System, Ghazvininejad et al, 2017 http://www.aclweb.org/anthology/P17-4008

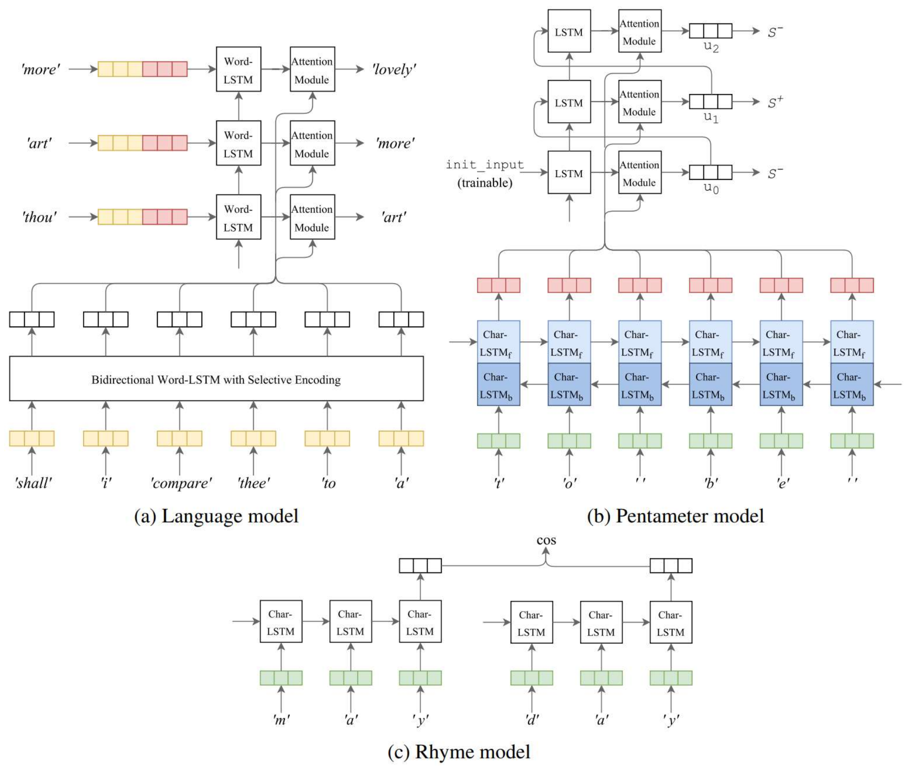
Poetry generation: Deep-speare

A more end-to-end approach to poetry generation (Lau et al)

Three components:

- language model
- pentameter model
- rhyme model

…learnt jointly as a multi-task learning problem



(a) Language model

(b) Pentameter model

(c) Rhyme model

Deep-speare: A joint neural model of poetic language, meter and rhyme, Lau et al, 2018
http://aclweb.org/anthology/P18-1181

Poetry generation: Deep-speare

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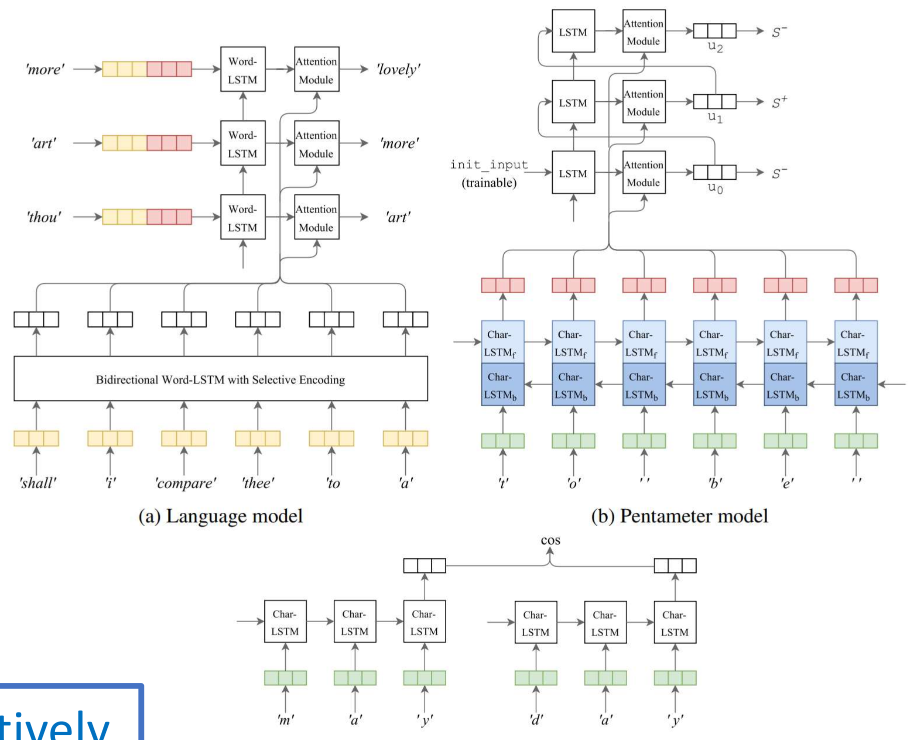


(a) Language model (b) Pentameter model (c) Rhyme model

Authors find that meter and rhyme is relatively easy to get right but the generated poems fall short on "emotion and readability"

Deep-speare: A joint neural model of poetic language, meter and rhyme, Lau et al, 2018

http://aclweb.org/anthology/P18-1181

Review generation

Product Reviews Generation

- The reviews are used to express opinions for different aspects of products, and have a wide variety of writing styles and different polarity strengths.
- Traditional methods for reviews generation rely extensively on hand-crafted rules and predefined templates.

Pattern-based Reviews Generation

- 捉妖记II

创意，音乐，画风，音乐和场面都很不错，认同，场面复古，效果尚可
值得推荐，井柏然演的好,越来越喜欢李宇春了
场面制作真心给力，也讲明白了。一二三四，刚好给四星。第一感觉，画面制作真心给力，无论是画工还是叙事还是节奏还是配乐都太棒，让观众身临其境
梁朝伟咋会接这么囧的片子
喜欢白百何又喜欢李宇春~
很漂亮的画面，配音及插曲都很不错，很漂亮的场面，特效很好，电影的配乐意外的好，镜头感
场景渲染，人物造型，色彩运用还有剪辑特别好，细节各方面都比较到位。
线索清晰，环节紧密，叙事紧凑，而且很正能量，很多人不喜欢音效，BGM，情节，这就是中国的东西！
冲着周迅和梁朝伟，还有作者麦家
还有是真的发现梁朝伟好会演戏阿
场景渲染，人物就有多水，色彩，细节各方面都比较到位。角度有多美，音乐很好听，主角内心独白有点幽默，这是我们的骄傲！配乐，动作，特效极富美感，剧结构奏推的平缓，没有绝对的主角和身份，人物
白百何脱发有点严重……
井柏然贡献了一把好演技
喜欢白百何，没有理由,真心觉得李宇春这次太进入角色，真的好
从此爱井柏然
场景渲染，人物造型，第一人称视点色彩总有些出戏。细节各方面都比较到位。不错看~别的不说，好久没有这么入戏的看完一部电影了。随着摄像机的镜头，调度服装布景各方面都很不错，简洁有力！
最近对白百何好感度颇高！是个敬业的家伙,李宇春的颜值是真的高
天王和井柏然好棒
还可以呢，还有就是音乐太棒了，但十分感人！画面很美！音乐也很不错，场面没的说，挺真实震撼的︶，四星，剧情紧凑，场面没的说，虽然有一些剧情上的硬伤，但现在的片子里，算诚意了类型题材的电影
亮点是李宇春演技,为了井柏然看的
挺好看的，梁朝伟的演技爆棚啊！
五星给白百何神演技
井柏然演技太棒了,李宇春真棒。片子不错。
终于看过了，节奏上有待改进，演技上有待改进，一颦一笑，场面，也在充分的展示人物感情
看完你会爱上白百何的,梁朝伟我爱你
白百何挺出彩的,李宇春演技赞
场面，而且很多景但是后期合成的，演员真的厉害。
特效没的说，推荐，色调和节奏感真好，飘忽的美术对凸显人物起到很好作用，怎么舒服怎么来。
网游背景加真人出演的设定一开始就想让人吐槽...井柏然白发造型杀马特审美无剧情可言台词生硬不过特效ok而且好多场景是在我交拍的233
井柏然的演技非常自然，他笑观众能跟着笑，他哭也能带动观众的情绪
要剧情有剧情场面有场面音效也很不错，算是一部诚意之作吧。要剧情有剧情场面有场面配乐也很不错，前段的节奏干净。四星是给画面的，让人不禁浮起了最为美妙的时候，bgm赞赞赞，四星是给镜头的
其实真的还行，角度还可以，场面还可以，情节比较紧凑，演技紧凑，我们看得也开心，人物给力。赞一个，都很细腻
第一人称视点视角总有些出戏。很喜欢。第一人称视点视角总有些出戏。拍的也好看，选角一流。
做一生梁朝伟的脑残粉,井柏然很惊艳！演技得到表现！

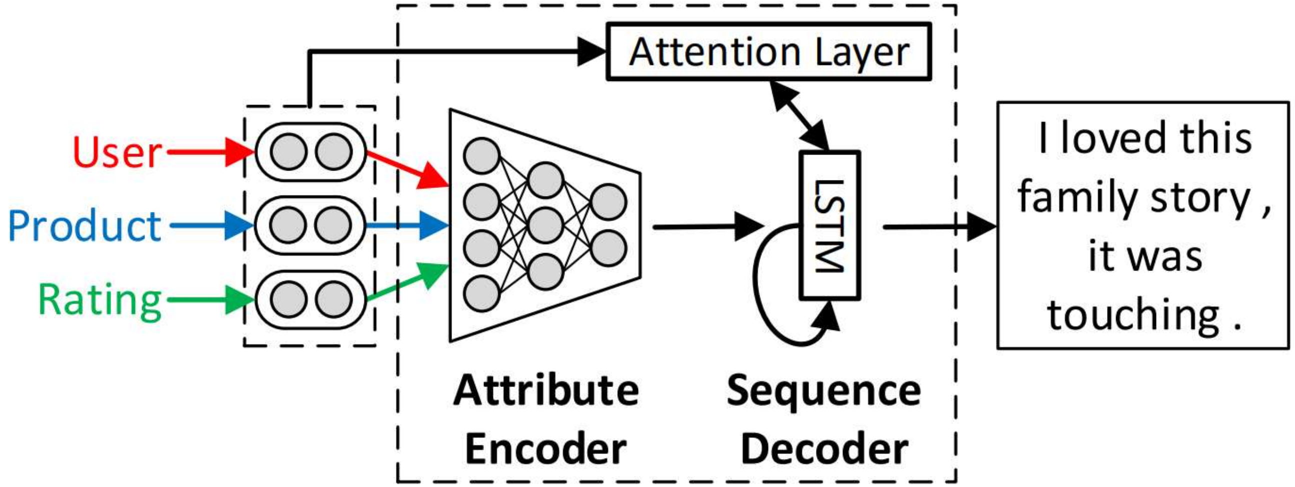
Pattern-based Reviews Generation

- 红海行动

片子是好片子，满足心里的某些叙事。。音效也不够流畅华丽，BGM，情节，这就是中国的东西！故事好玩视角与情节相比缺乏想象力，不过选这个题材还是蛮好看的……大场面令人叹为观止，美术造诣不一般呐
张译演的好啊,杜江演戏还是很有演技的
场面很美，具有深刻的教育意义。演员的表演也很用心，比起一些导演随便制作的烂片，强太多太多了。最喜欢这种结构发散，角度很美，最后一段又不失感人~音效，故事太散，人物平淡，剧情，音效，语言，动作都
情节很紧凑，张译依旧给力,杜江演的真棒
海清的角色很精彩,杜江的演技巅峰了么
无论是画工还是人物还是节奏还是配乐都太棒，细腻，情节不错，叙事也不错！失败的故事。
场面真TM华丽，音乐也不错，内容深刻。意料之外视角给的真好，拍的也好看，选角一流。意料之外视角给的真好，拍的也好看，选角一流。
镜头精细、配乐有煽动性。特别喜欢，很有意思。其实真的还行，角度精细、配乐有煽动性。特别喜欢，電影的畫面，光影，角度很精美，色彩都很漂亮，就像一幅幅活動的PS效果濃重的照片
不得不说，超出预期太多，画面、台词、表演、配乐，拥有现代感觉的结合电影没有将故事表现。
音效很赞，故事太散，人物平淡，画面效果精良，音乐很好听，这是我们的骄傲！配乐，动作，打斗特效明快点到为止我觉得很久没听过这么好听的故事了。
干净的画面，台词音乐超棒，虽然没有大场面，赞爆了！台词服装布景各方面都很不错，都是前后文略显生硬迷失在他的故事。
海清演的好啊,杜江很出彩
電影的畫面，光影，色彩和角度真心不错，色彩都很漂亮，就像一幅幅活動的PS效果濃重的照片，色彩和场面真心不错，值得到电影院看的好剧，为了导演去看的片子，美术晃得我很喜欢，有些镜头很清爽不错，剧情
张译，不错的我爱的海清！
杜江演的真好……为张译点一百个赞
满满的治愈，音乐是亮点姑娘变公主被王子爱上的故事但不得不承认人家玩得很好。
镜头太美丽了，都直击心扉。真是厉害啊！结构新奇情节紧凑，是演技……
情节节奏也还不错。角色出彩。我喜欢的风格场面渲染的没的说，一点也不无聊，很精彩生动，插曲很棒，国产片里特效和镜头算震撼的了！除了结尾太仓促。音效棒，算是一部诚意之作吧。
打斗的动作流畅性、场面和人物制作都很好，但是叙事有点拖沓
杜江必须怒赞啊,张译的出演真是赞不绝口。
结构新奇情节紧凑，不然给个五星。电影情节不拖沓，演员入戏，其实好电影真的很简单，配乐太对胃了！场面太美了，不需要打牌，精致的细节能和观众产生些许共鸣即是好电影。
满分给张译演的杀手
挺喜欢看海清的电影～
剧情有点假，人物还可以，很漂亮的画面，这样的节奏、氛围、美术、灯光
张译很好看,做一生杜江的脑残粉
在进行场面展开时但是叙事有点拖沓，视觉效果还是很好的，在进行场面展开时
角度浓重，生活更加多姿多彩。。角度和配乐都很完美，生活更加多姿多彩。。最想说的就是，特效很震撼
音乐与镜头的配合，故事悬疑、结局悬念。总体上来说算是国产佳片。整部电影节奏感把握的很好，经典枪战特效，太精彩了！画面没得说，剧情略薄弱，剧本很紧凑，叙事很扎实，使用的服装道具很好啊

Neural Reviews Generation

- Dong, Li, et al presents neural approach to generating product reviews
- The model learns to encode attributes into vectors, and then uses recurrent neural networks based on long short-term memory (LSTM) units to generate reviews
- An attention layer is used to learn soft alignments between attributes and generated words



Learning to Generate Product Reviews from Attributes, dong li et al 2017 https://www.aclweb.org/anthology/E17-1059.pdf

NLG evaluation

Automatic evaluation metrics for NLG

Word overlap based metrics (BLEU, ROUGE, METEOR, F1, etc.)

- We know that they're not ideal for machine translation
- They're even worse for summarization, which is more open-ended than machine translation
  - Unfortunately, ROUGE also rewards extractive summarization systems more than abstractive systems
- And they're even worse for dialogue, which is more open-ended that summarization.
  - Similarly for e.g. story generation

Automatic evaluation metrics for NLG

Word overlap metrics are not good for dialogue

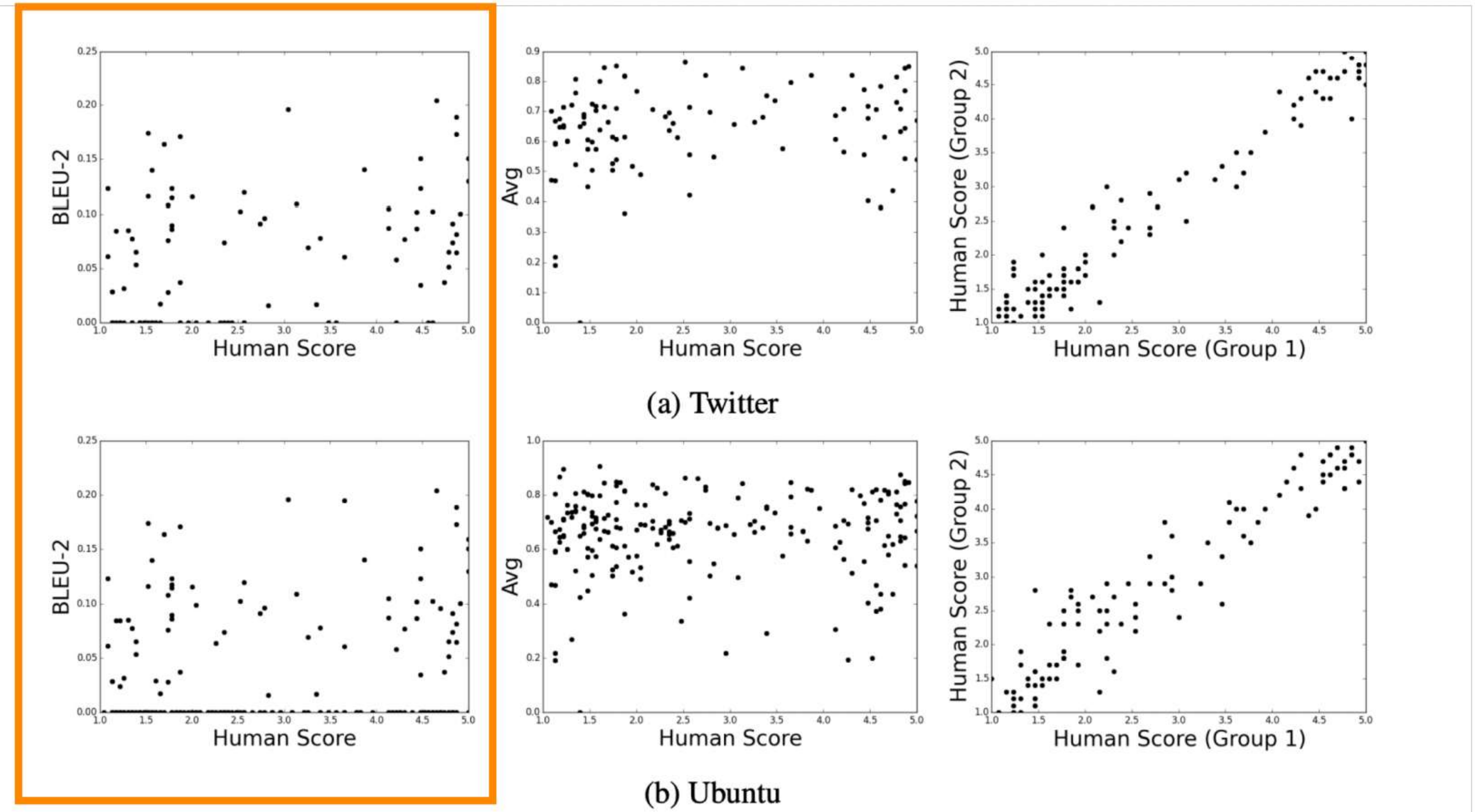


Figure 1: Scatter plots showing the correlation between metrics and human judgements on the Twitter corpus (a) and Ubuntu Dialogue Corpus (b). The plots represent BLEU-2 (left), embedding average (center), and correlation between two randomly selected halves of human respondents (right).

How NOT To Evaluate Your Dialogue System: An Empirical Study of Unsupervised Evaluation Metrics for Dialogue Response Generation, Liu et al, 2017 [*https://arxiv.org/pdf/1603.08023.pdf*](https://arxiv.org/pdf/1603.08023.pdf)

Automatic evaluation metrics for NLG

What about perplexity?

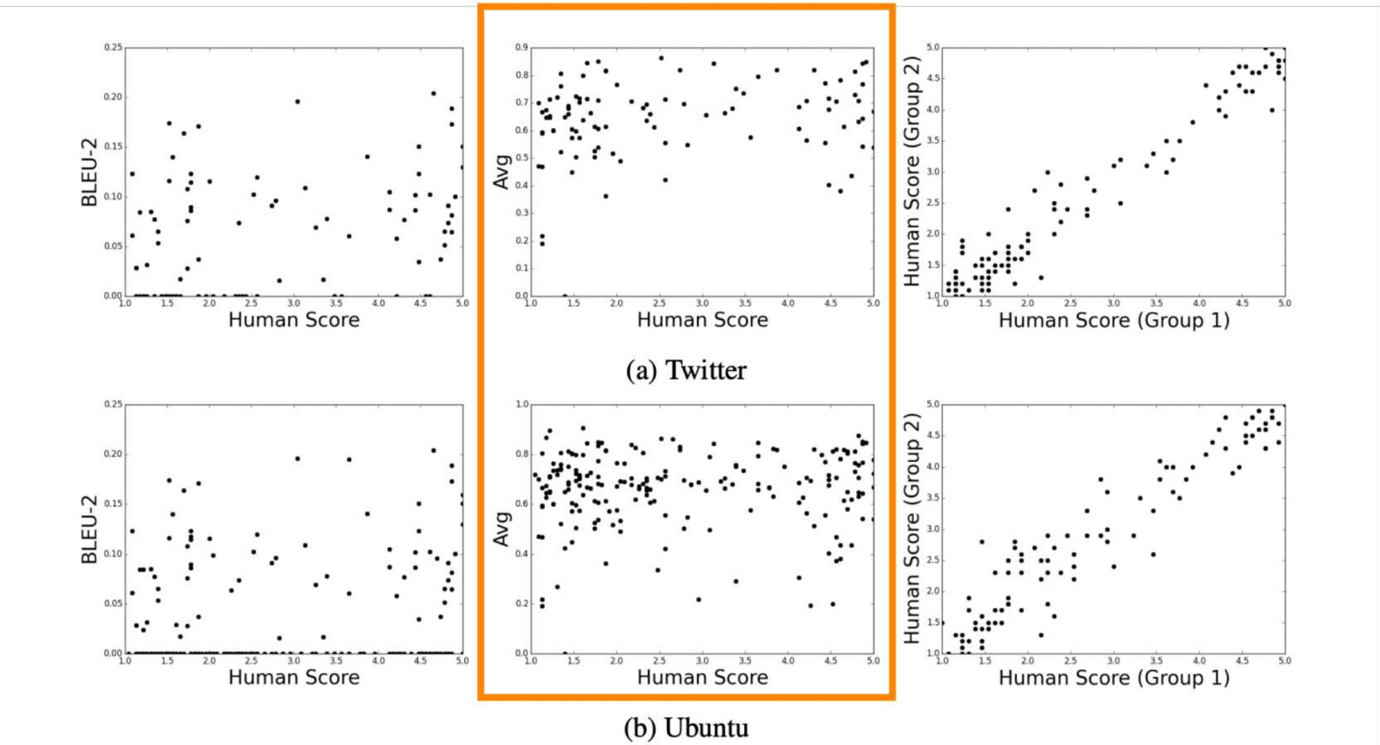
- Captures how powerful your LM is, but doesn't tell you anything about generation (e.g. if your decoding algorithm is bad, perplexity is unaffected)

Word embedding based metrics?

- Main idea: compare the similarity of the word embeddings (or average of word embeddings), not just the overlap of the words themselves. Captures semantics in a more flexible way.
- Unfortunately, still doesn't correlate well with human judgments for open-ended tasks like dialogue.

Automatic evaluation metrics for NLG

Word embedding based metrics still doesn't correlate well with human judgments



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Automatic evaluation metrics for NLG

We have no automatic metrics to adequately capture overall quality (i.e. a proxy for human quality judgment).

But we can define more focused automatic metrics to capture particular aspects of generated text:

- Fluency (compute probability w.r.t. well-trained LM)
- Correct style (prob w.r.t. LM trained on target corpus)
- Diversity (rare word usage, uniqueness of n-grams)
- Relevance to input (semantic similarity measures)
- Simple things like length and repetition
- Task-specific metrics e.g. compression rate for summarization

Though these don't measure overall quality, they can help us track some important qualities that we care about.

Human evaluation

- Human judgments are regarded as the **gold standard**
- Of course, we know that human eval is slow and expensive
- ...but are those the only problems?
- Supposing you do have access to human evaluation:
  Does human evaluation solve all of your problems?

- **No!**
- Conducting human evaluation effectively is very difficult
- Humans:
  - are inconsistent
  - can be illogical
  - lose concentration
  - misinterpret your question
  - can't always explain why they feel the way they do

Possible new avenues for NLG eval?

- Corpus-level metrics
  - Should an eval metric be applied to each example in the test set independently, or a function of the whole corpus?
  - e.g. if a dialogue model always gives the same generic answer to every example in the test set, it should be penalized
- Eval metrics that measure the diversity-safety tradeoff
  - Human eval for free
  - Gamification: make the task (e.g. talking to a chatbot) fun, so humans provide supervision and implicit evaluation for free
- Adversarial discriminator as an evaluation metric
  - Test whether the NLG system can fool a discriminator which is trained to distinguish human text from artificially generated text

Exciting current trends in NLG

- Incorporating discrete latent variables into NLG
  - May help with modeling structure in tasks that really need it, like storytelling, task-oriented dialogue, etc
- Alternatives to strict left-to-right generation
  - Parallel generation, iterative refinement, top-down generation for longer pieces of text
- Alternative to maximum likelihood training with teacher forcing
  - More holistic sentence-level (rather than word-level) objectives

Neural NLG community is rapidly maturing

- During the early years of NLP + Deep Learning, community was mostly transferring successful NMT methods to NLG tasks.
- Now, increasingly more inventive NLG techniques emerging, specific to non-NMT generation settings.
- Increasingly more (neural) NLG workshops and competitions, especially focusing on open-ended NLG:
  - NeuralGen workshop
  - Storytelling workshop
  - Alexa challenge
  - ConvAI2 NeurIPS challenge
- These are particularly useful to organize the community, increase reproducibility, standardize eval, etc.
- **The biggest roadblock for progress is eval !**

Tips for working in NLG

1. The more open-ended the task, the harder everything becomes.
   - Constraints are sometimes welcome!
2. Aiming for a specific improvement can be more manageable than aiming to improve overall generation quality.
3. If you’re using a LM for NLG: improving the LM (i.e. perplexity) will most likely improve generation quality.
   - ...but it's not the only way to improve generation quality.
4. Look at your output, a lot

Tips for working in NLG

5. You need an automatic metric, even if it's imperfect.
   - You probably need several automatic metrics.
6. If you do human eval, make the questions as focused as possible.
7. Reproducibility is a huge problem in today's NLP + Deep Learning, and a huger problem in NLG.
   - Please, publicly release all your generated output along with your paper!
8. Working in NLG can be very frustrating. But also very funny…